

Visión por Computador

Versión 2026

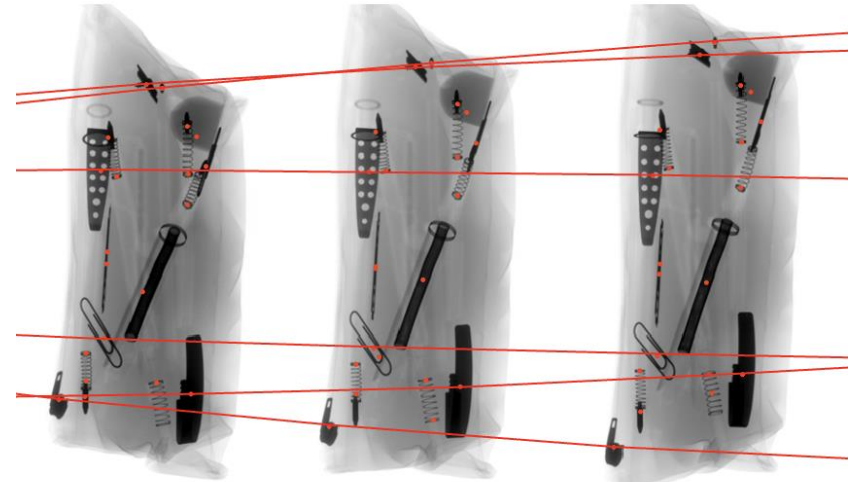


Múltiples Vistas

(CAPÍTULO 2 – CV02F)

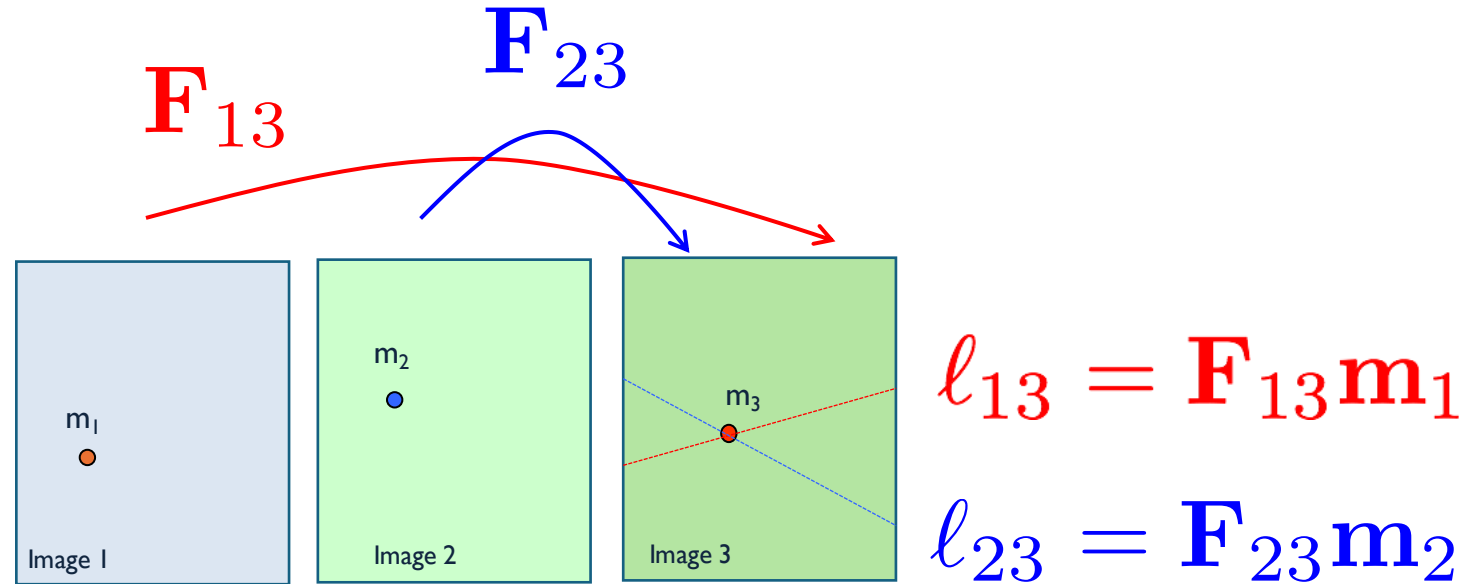
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Geometría Trifocal

Epipolar Geometry for 3 views:



$$\mathbf{F}_{13} = \text{fundamental}(\mathbf{A}, \mathbf{C})$$

$$\mathbf{F}_{23} = \text{fundamental}(\mathbf{B}, \mathbf{C})$$

Problem:

Given $\mathbf{m}_1, \mathbf{m}_2$ (and projection matrices $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$)
find $\mathbf{m}_3$

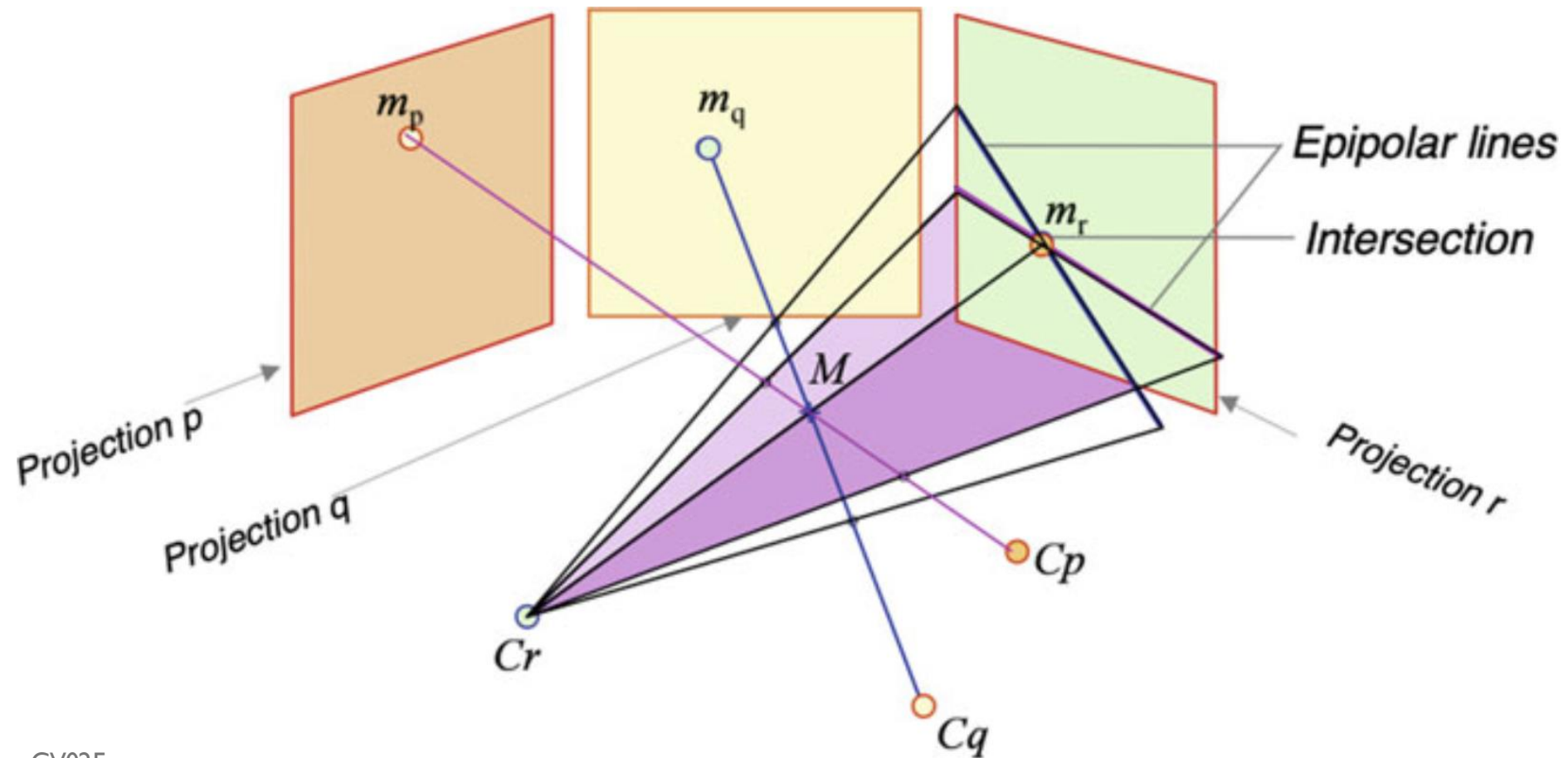
$$\mathbf{m}_3 = \ell_{13} \times \ell_{23}$$

$$\lambda_1 \mathbf{m}_1 = \mathbf{A}\mathbf{M}$$

$$\lambda_2 \mathbf{m}_2 = \mathbf{B}\mathbf{M}$$

$$\lambda_3 \mathbf{m}_3 = \mathbf{C}\mathbf{M}$$

$$\begin{cases} \lambda_p \mathbf{m}_p = \mathbf{P}_p \mathbf{M} := \mathbf{A} \mathbf{M} \\ \lambda_q \mathbf{m}_q = \mathbf{P}_q \mathbf{M} := \mathbf{B} \mathbf{M} \\ \lambda_r \mathbf{m}_r = \mathbf{P}_r \mathbf{M} := \mathbf{C} \mathbf{M} \end{cases}$$



$$\begin{cases} \lambda_p \mathbf{m}_p = \mathbf{P}_p \mathbf{M} := \mathbf{A} \mathbf{M} \\ \lambda_q \mathbf{m}_q = \mathbf{P}_q \mathbf{M} := \mathbf{B} \mathbf{M} \\ \lambda_r \mathbf{m}_r = \mathbf{P}_r \mathbf{M} := \mathbf{C} \mathbf{M} \end{cases} \quad \text{---} \downarrow$$

$$\underbrace{\begin{bmatrix} \mathbf{a}_1 & x_p & 0 & 0 \\ \mathbf{a}_2 & y_p & 0 & 0 \\ \mathbf{a}_3 & 1 & 0 & 0 \\ \mathbf{b}_1 & 0 & x_q & 0 \\ \mathbf{b}_2 & 0 & y_q & 0 \\ \mathbf{b}_3 & 0 & 1 & 0 \\ \mathbf{c}_1 & 0 & 0 & x_r \\ \mathbf{c}_2 & 0 & 0 & y_r \\ \mathbf{c}_3 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{G}} \underbrace{\begin{bmatrix} \mathbf{M} \\ -\lambda_p \\ -\lambda_q \\ -\lambda_r \end{bmatrix}}_{\mathbf{v}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad (3.74)$$

where $\mathbf{a}_i$, $\mathbf{b}_i$ and $\mathbf{c}_i$ denotes the i th row of matrix $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ respectively. If m_p , m_q and m_r are corresponding points, then there must be a nontrivial solution for $\mathbf{v}$. It follows that the rank of the 9×7 matrix $\mathbf{G}$ must be at most 6. In other words, all 7×7 submatrices have vanishing determinants.

$$\begin{cases} D_1 = (x_r \mathbf{T}^{13} - x_r x_q \mathbf{T}^{33} + x_q \mathbf{T}^{31} - \mathbf{T}^{11}) \mathbf{m}_p = 0 \\ D_2 = (y_r \mathbf{T}^{13} - y_r x_q \mathbf{T}^{33} + x_q \mathbf{T}^{32} - \mathbf{T}^{12}) \mathbf{m}_p = 0 \\ D_3 = (x_r \mathbf{T}^{23} - x_r y_q \mathbf{T}^{33} + y_q \mathbf{T}^{31} - \mathbf{T}^{21}) \mathbf{m}_p = 0 \\ D_4 = (y_r \mathbf{T}^{23} - y_r y_q \mathbf{T}^{33} + y_q \mathbf{T}^{32} - \mathbf{T}^{22}) \mathbf{m}_p = 0 \end{cases}, \quad (3.75)$$

where

$$\mathbf{T}^{jk} = [T_1^{jk} \ T_2^{jk} \ T_3^{jk}],$$

and

$$T_i^{jk} = (-1)^{i+1} \left| \begin{array}{c} \sim \mathbf{a}_i \\ \mathbf{b}_j \\ \mathbf{c}_k \end{array} \right| = \quad \text{for } i, j, k = 1, 2, 3, \quad (3.76)$$

where $\sim \mathbf{a}_i$ means the matrix $\mathbf{A}$ without row i . The elements T_i^{jk} are called the *trifocal tensors* for the images p , q and r [10, 11]. For the canonical form $\mathbf{A} = [\mathbf{I} \mid \mathbf{0}]$, the trifocal tensors may be easily obtained by

$$T_i^{jk} = b_{ji} c_{k4} - b_{j4} c_{ki} \quad \text{for } i, j, k = 1, 2, 3. \quad (3.77)$$

The *reprojection* of m_r , i.e., the coordinates $\hat{x}_r$ and $\hat{y}_r$ obtained from the points m_p and m_q , may be simply estimated from the trilinearities (3.75):

$$\lambda \hat{\mathbf{m}}_r = (\mathbf{T}^1 - x_q \mathbf{T}^3) \mathbf{m}_p = (\mathbf{T}^2 - y_q \mathbf{T}^3) \mathbf{m}_p, \quad (3.78)$$

where λ is a scale factor, $\hat{\mathbf{m}}_r = [\hat{x}_r \ \hat{y}_r \ 1]^\top$, and $\mathbf{T}^j$ is a 3×3 matrix with the (k, i) -element equal to T_i^{jk} .

In practice, given two corresponding points $\mathbf{m}_p$ and $\mathbf{m}_q$, the third one $\mathbf{m}_r$ can be considered as the corresponding point in third view, if the Euclidean distance between $\mathbf{m}_r$ and its reprojection $\hat{\mathbf{m}}_r$ is smaller than a small number ε :

$$d_3 = \|\hat{\mathbf{m}}_r - \mathbf{m}_r\| < \varepsilon. \quad (3.79)$$

Example

Figure F shows a transparent bag containing various items. A blue arrow points to a red dot on the bag, labeled "INPUT I". The items inside the bag include a coiled cable, a small electronic device, and a black bag.

Figure q



Figure r



Application: Bagagge Screening

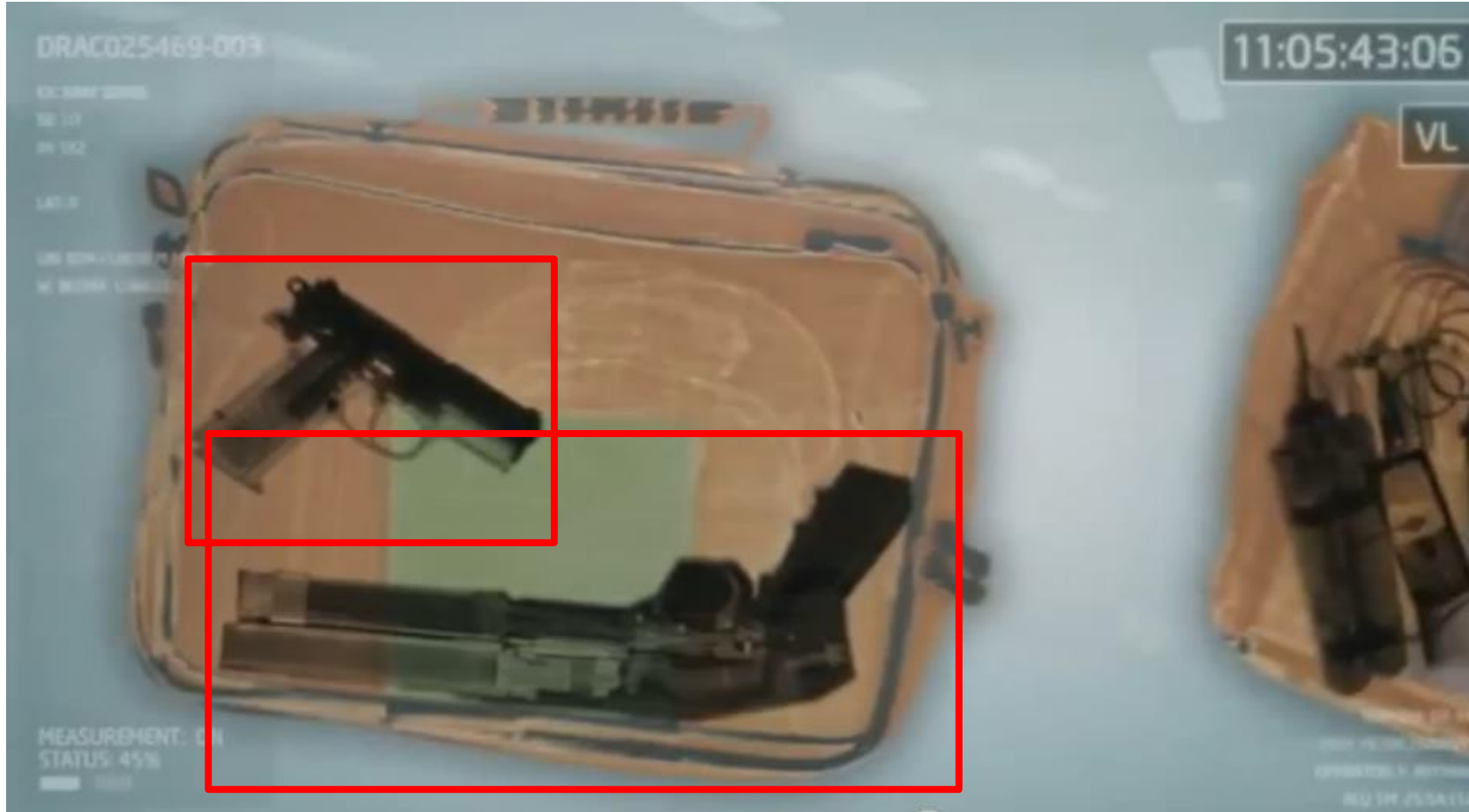
X-ray testing: **Baggage screening**

X-ray testing: **Baggage screening**



X-ray testing: **Baggage screening**

X-ray testing: **Baggage screening**



Our goal:

- 1) Detection of objects of interest
- 2) Views of each object in all frames



X-ray testing: **Baggage screening**

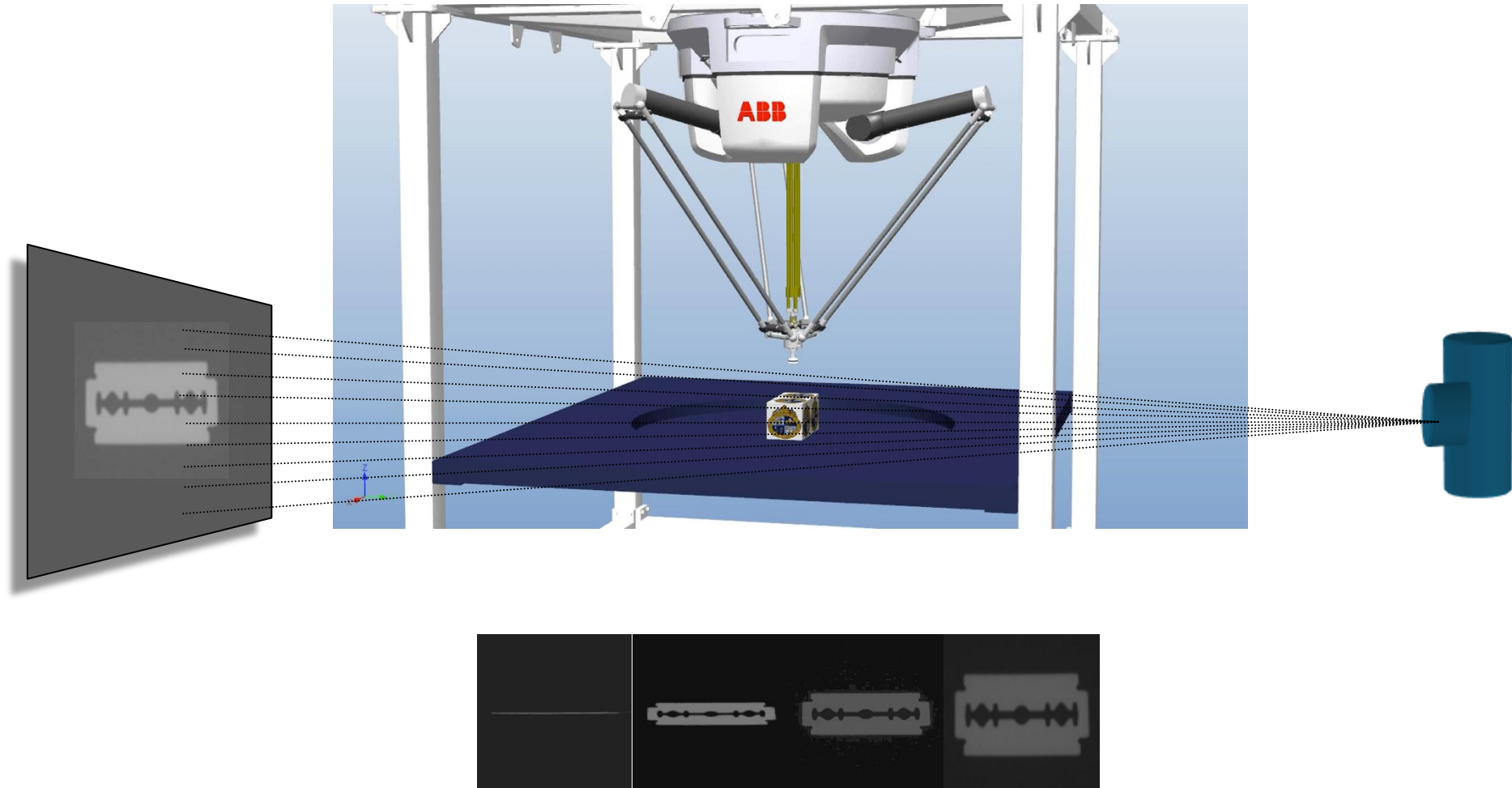
Key ideas of our methods:

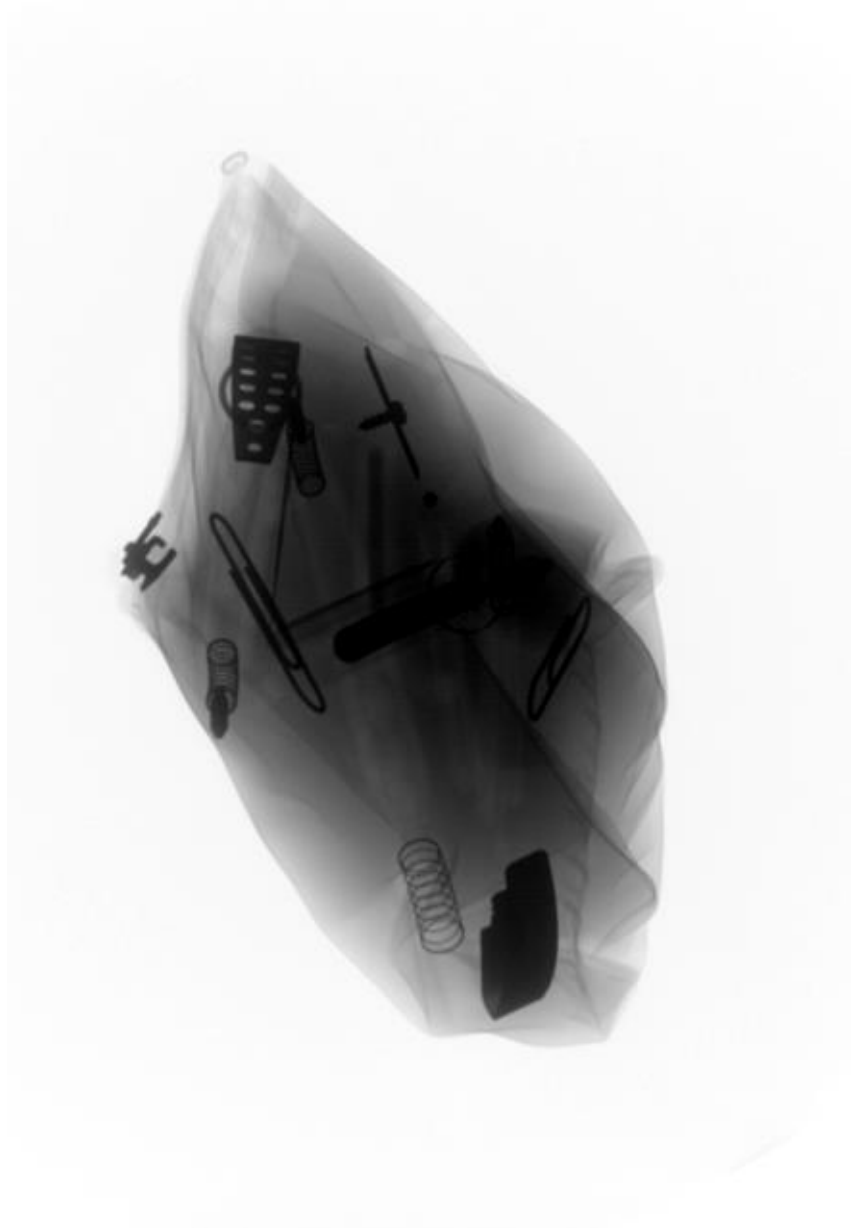
- **Computer vision**
- **Multiple views analysis**
- **Active vision**



X-ray Testing: Multiple view analysis

X-ray Testing: Multiple view analysis



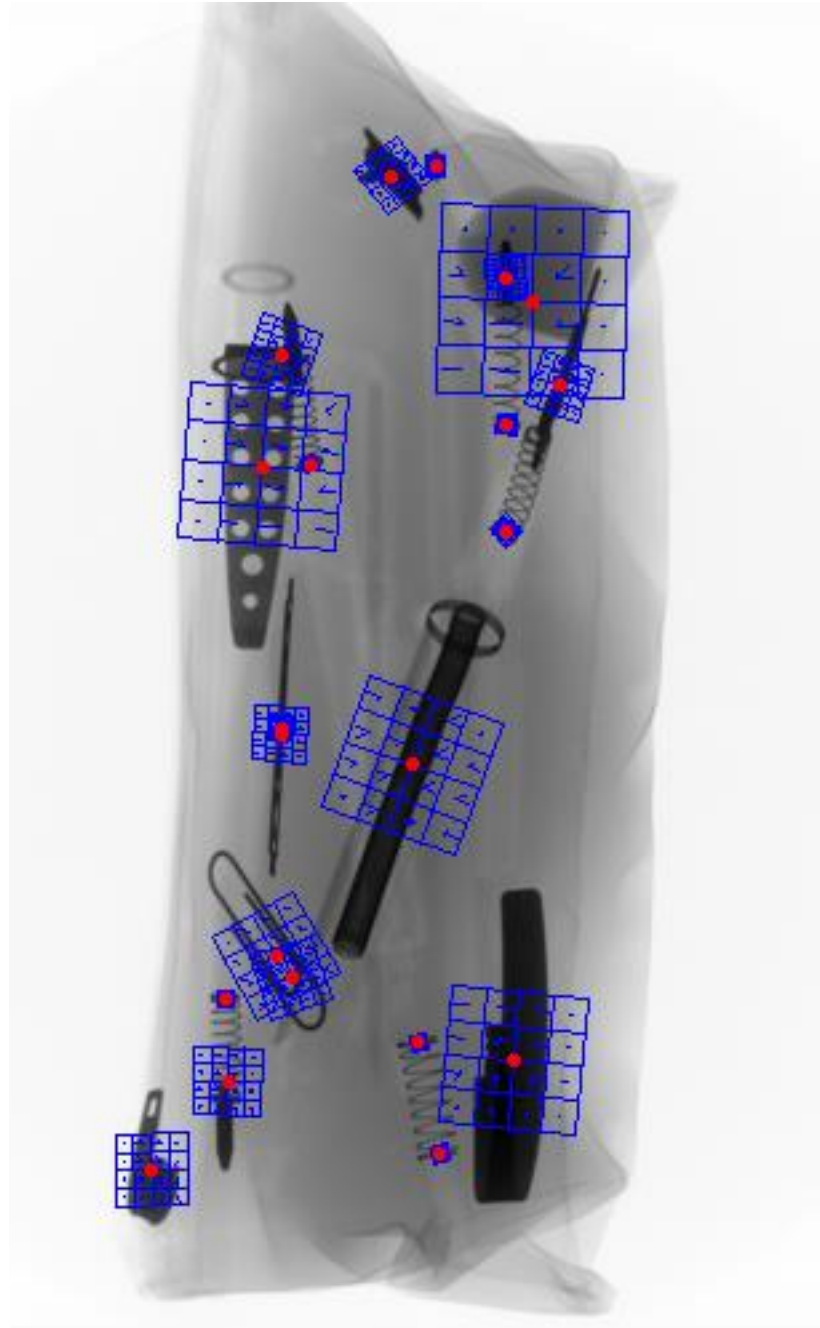




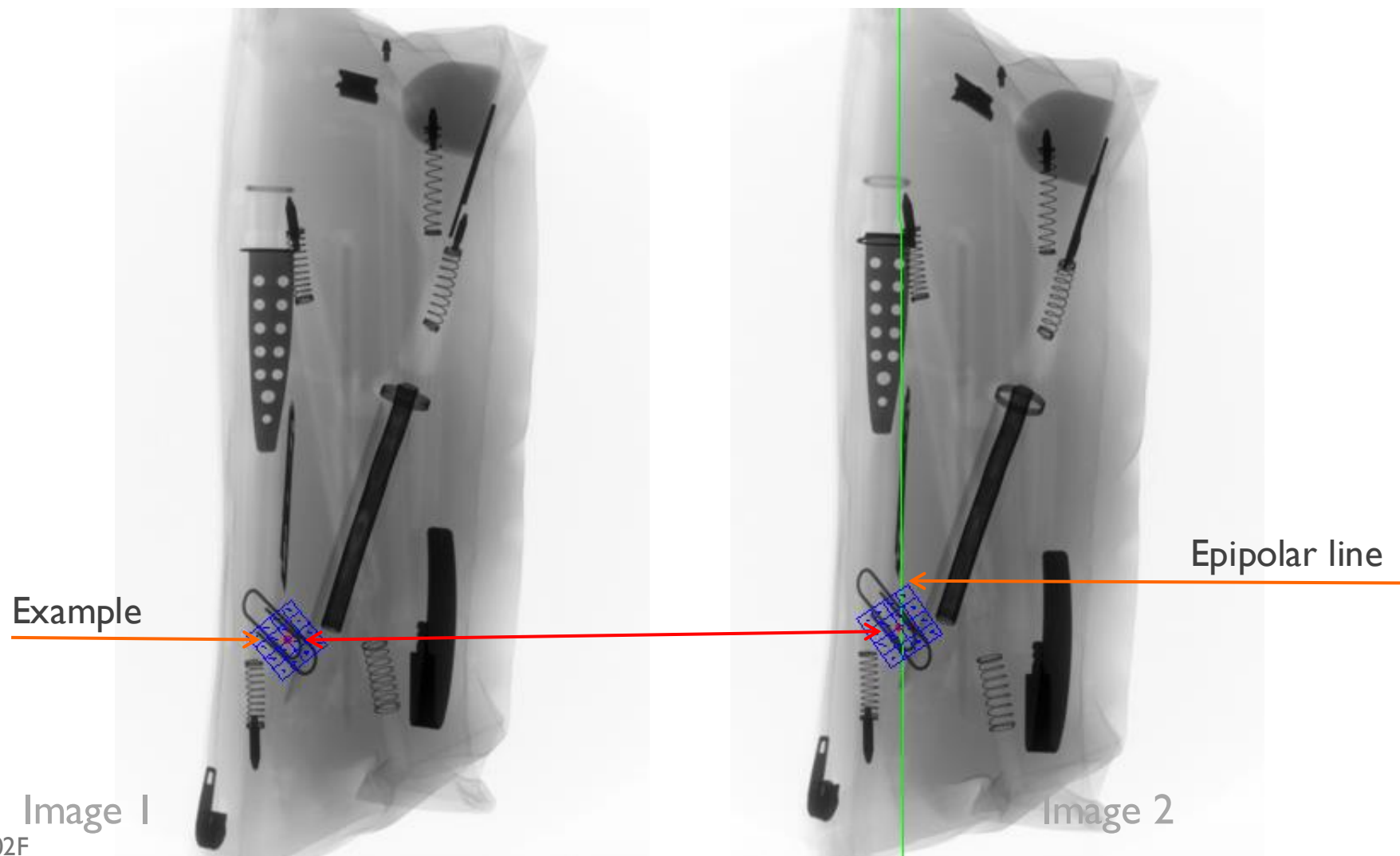




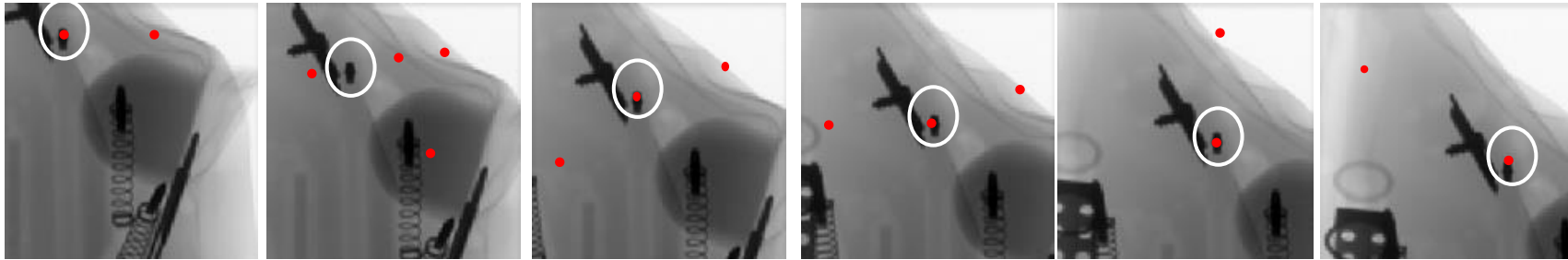
SIFT description for each
segmented region



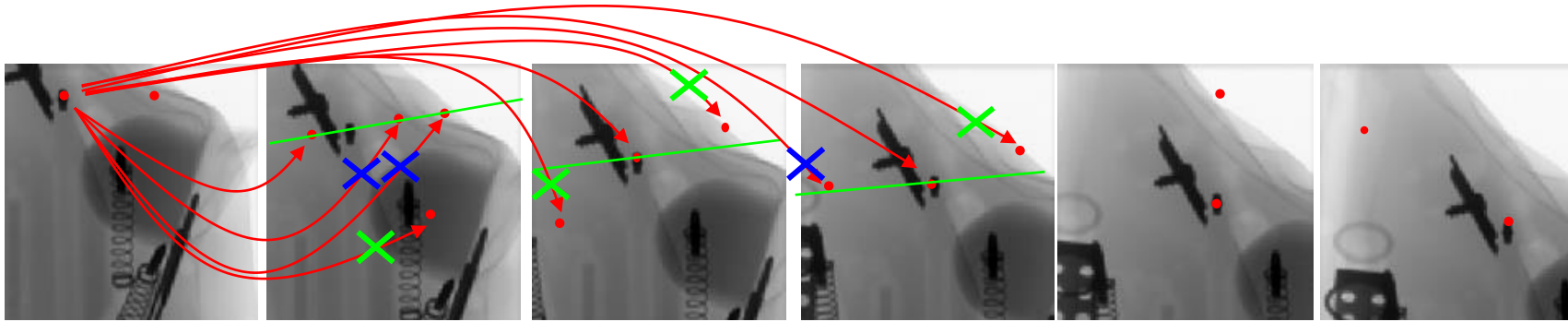
Tracking in 2 views



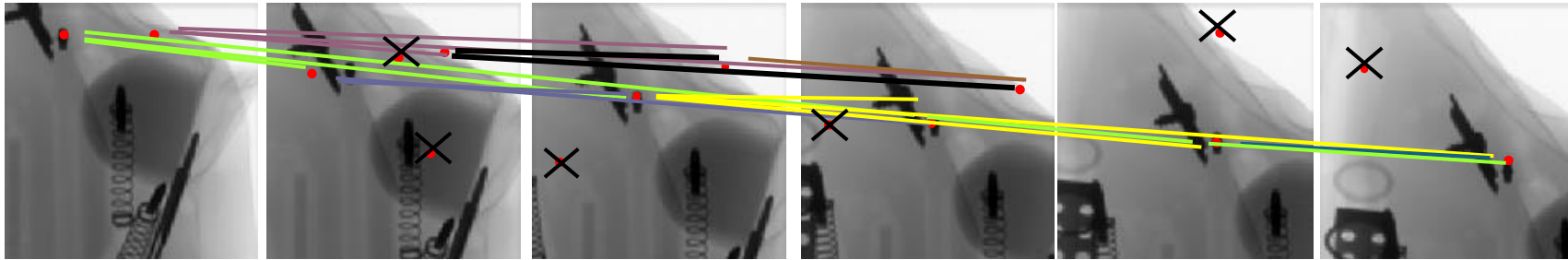
Tracking algorithm



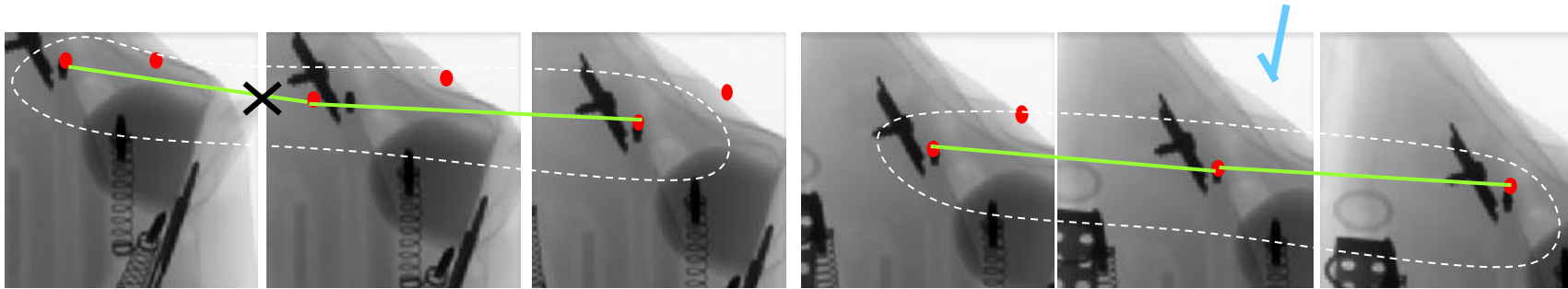
Tracking algorithm



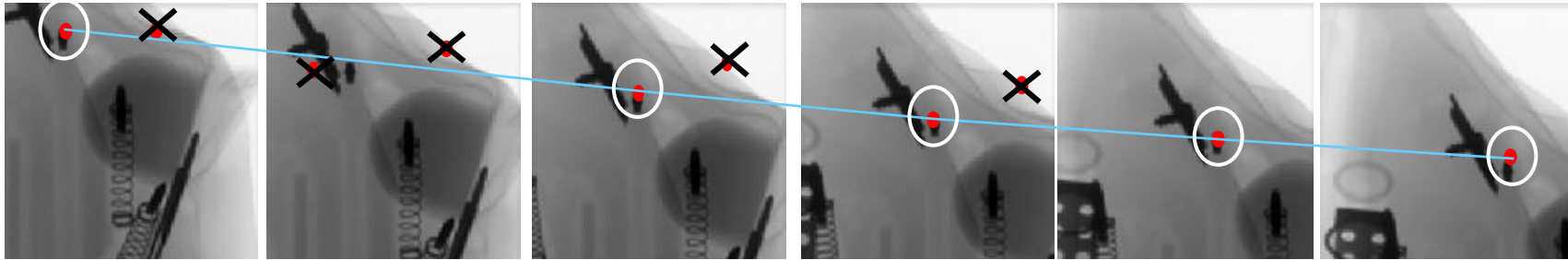
Tracking algorithm



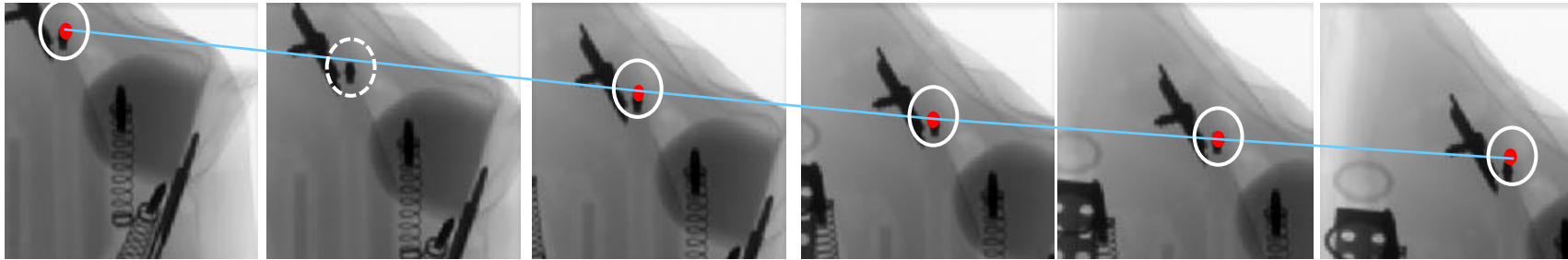
Tracking algorithm



Tracking algorithm

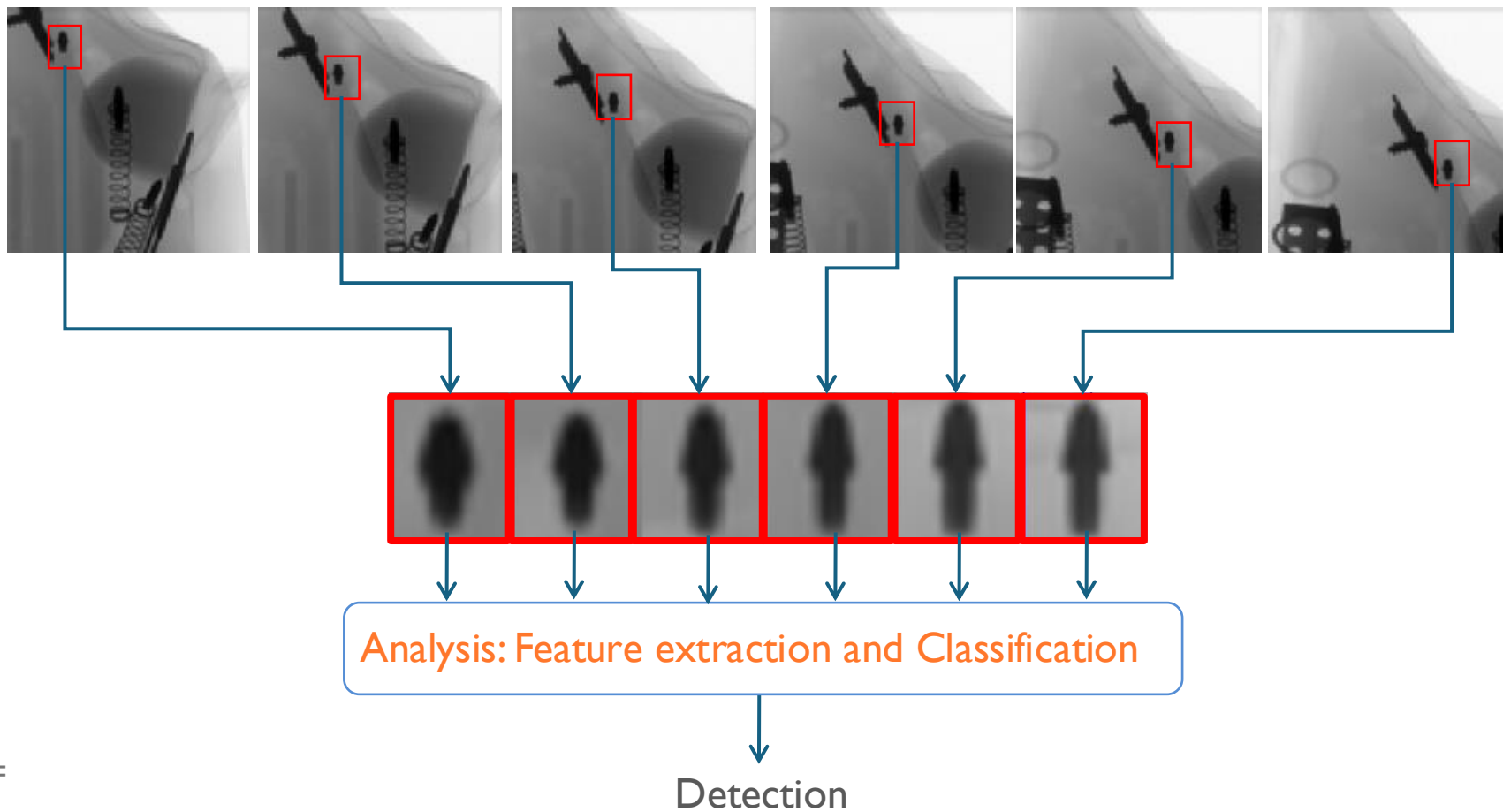


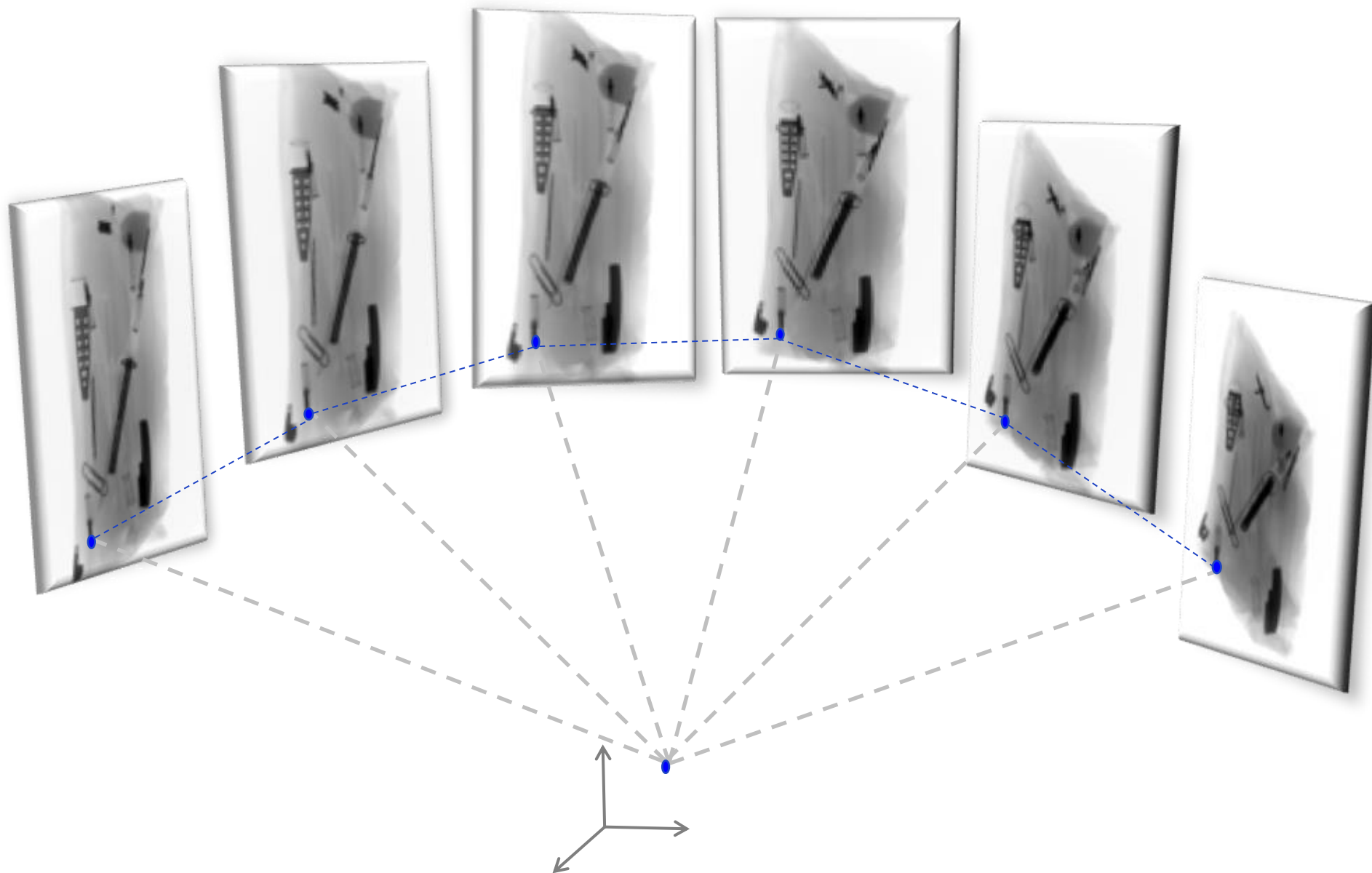
Tracking algorithm

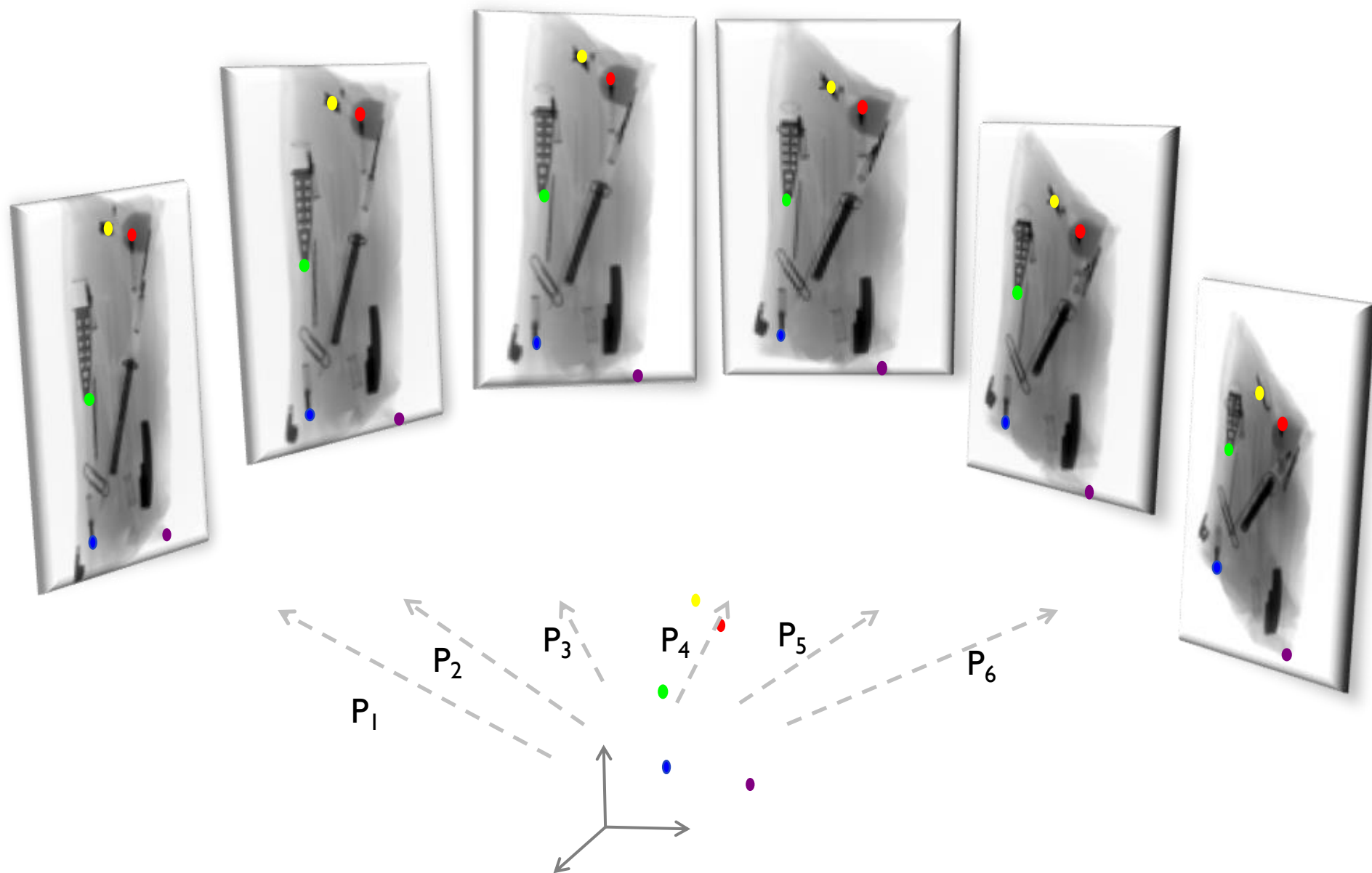


- It is not required that the object parts have to be segmented in all views because the missing views can be reconstructed.
- It is allowed to have false segmentations because they are filtered out.

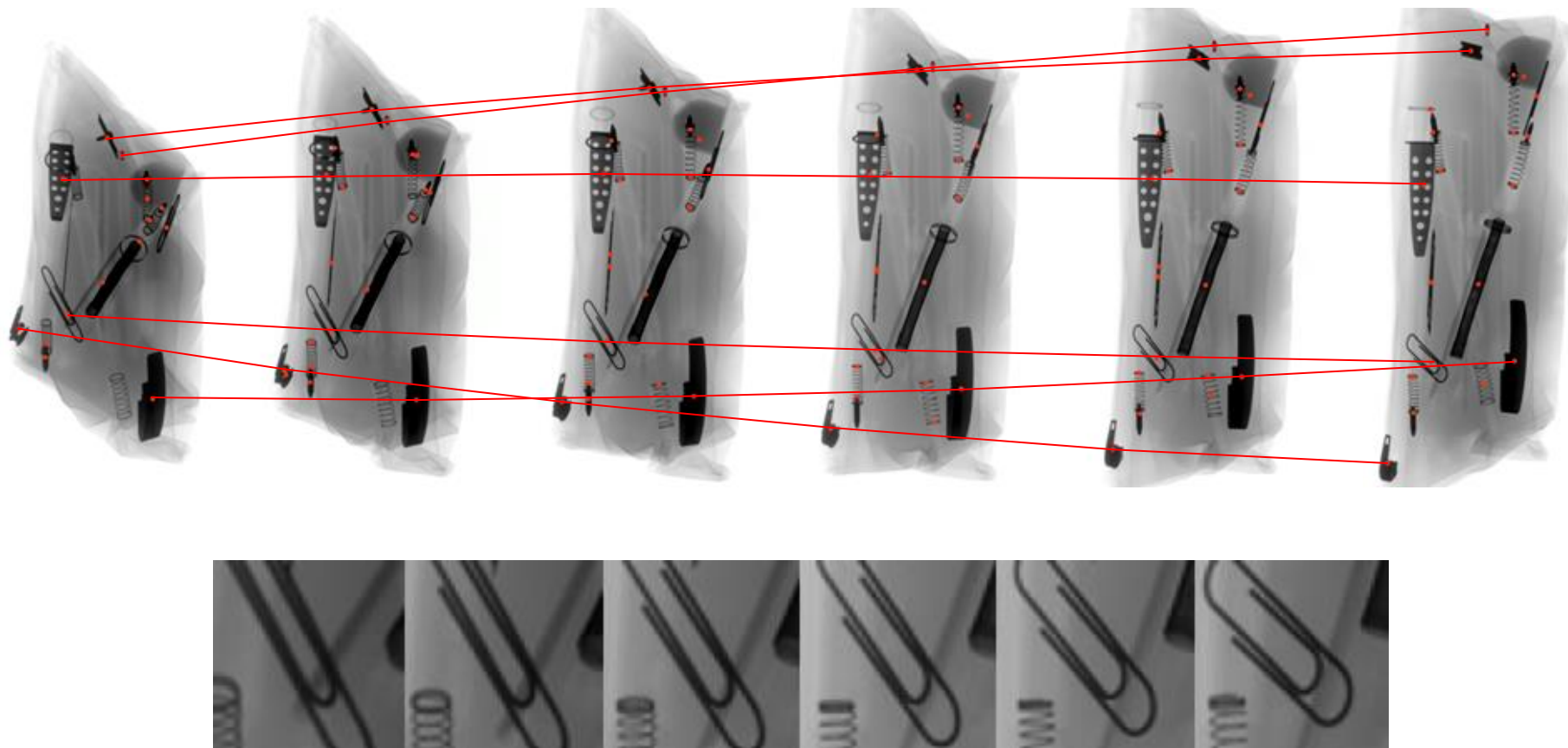
Tracking algorithm



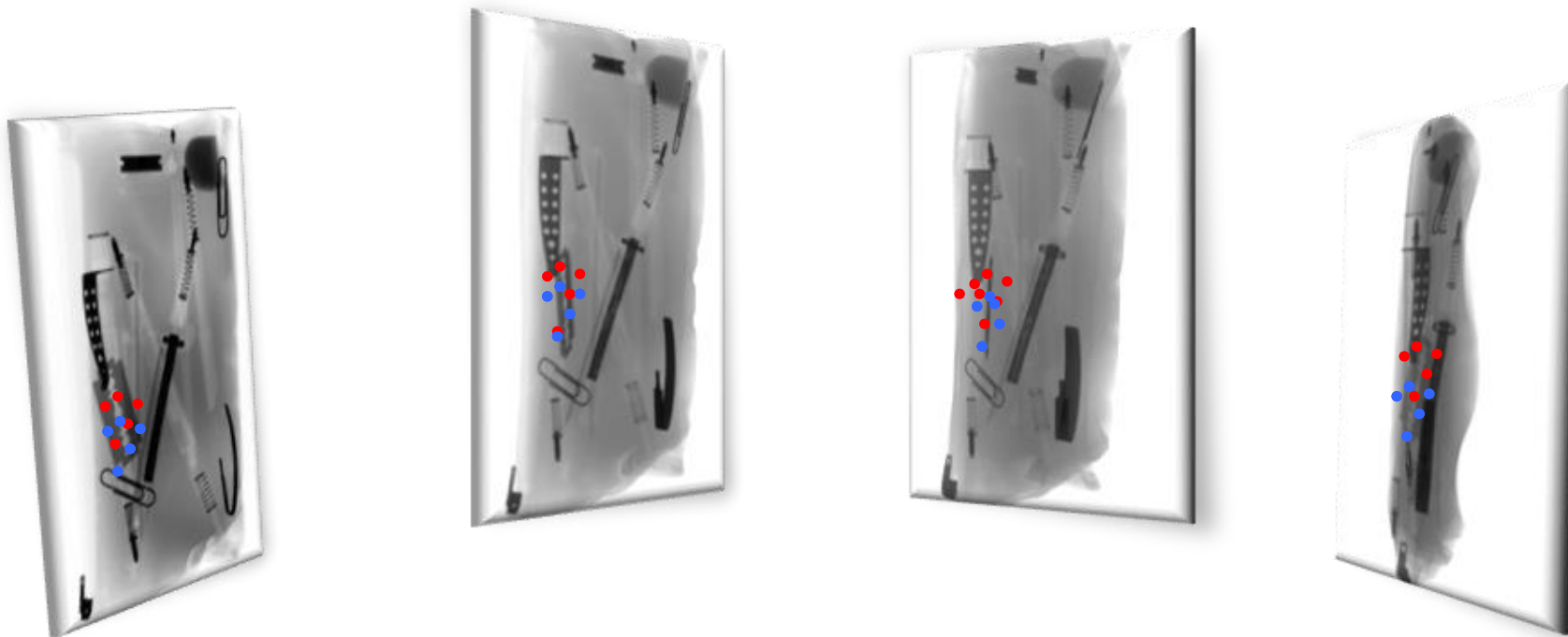


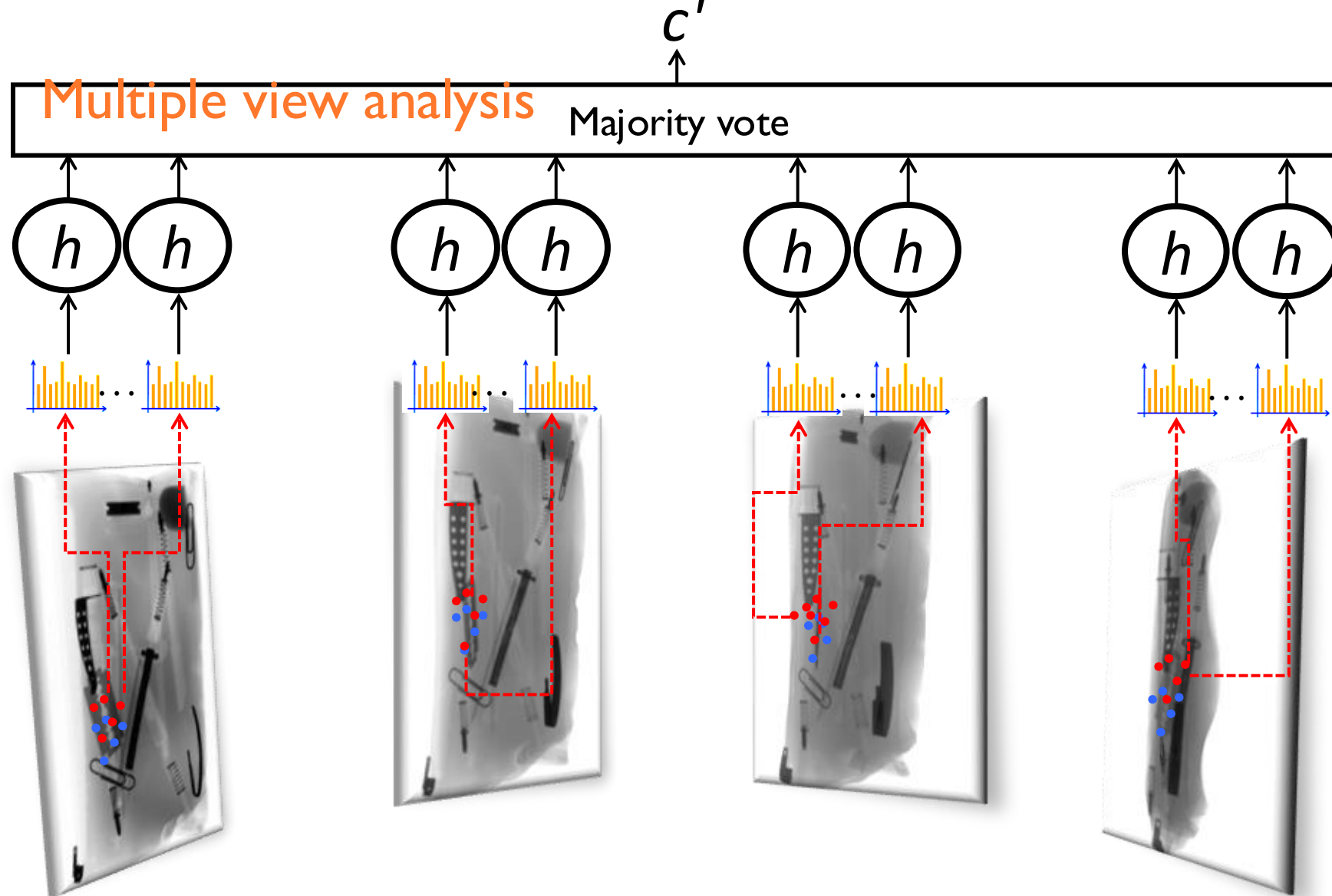


X-ray Testing: Multiple view analysis

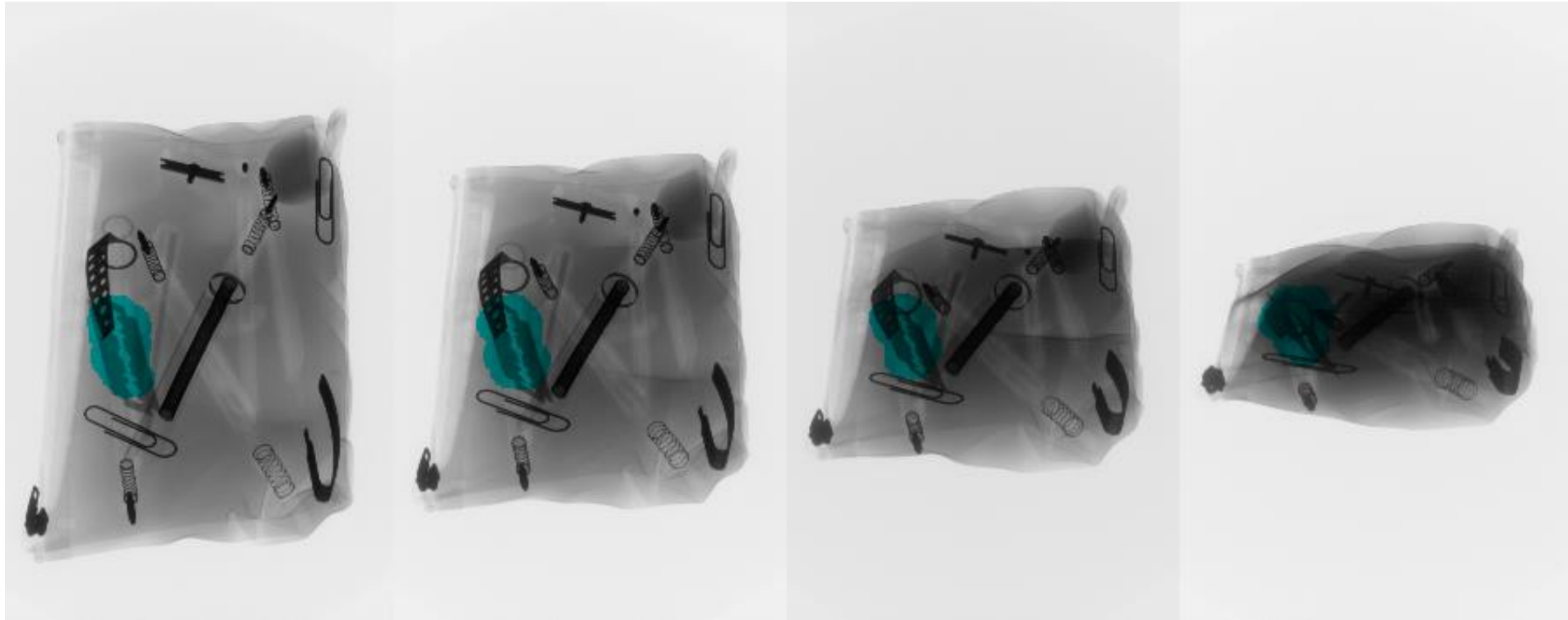


Multiple view analysis



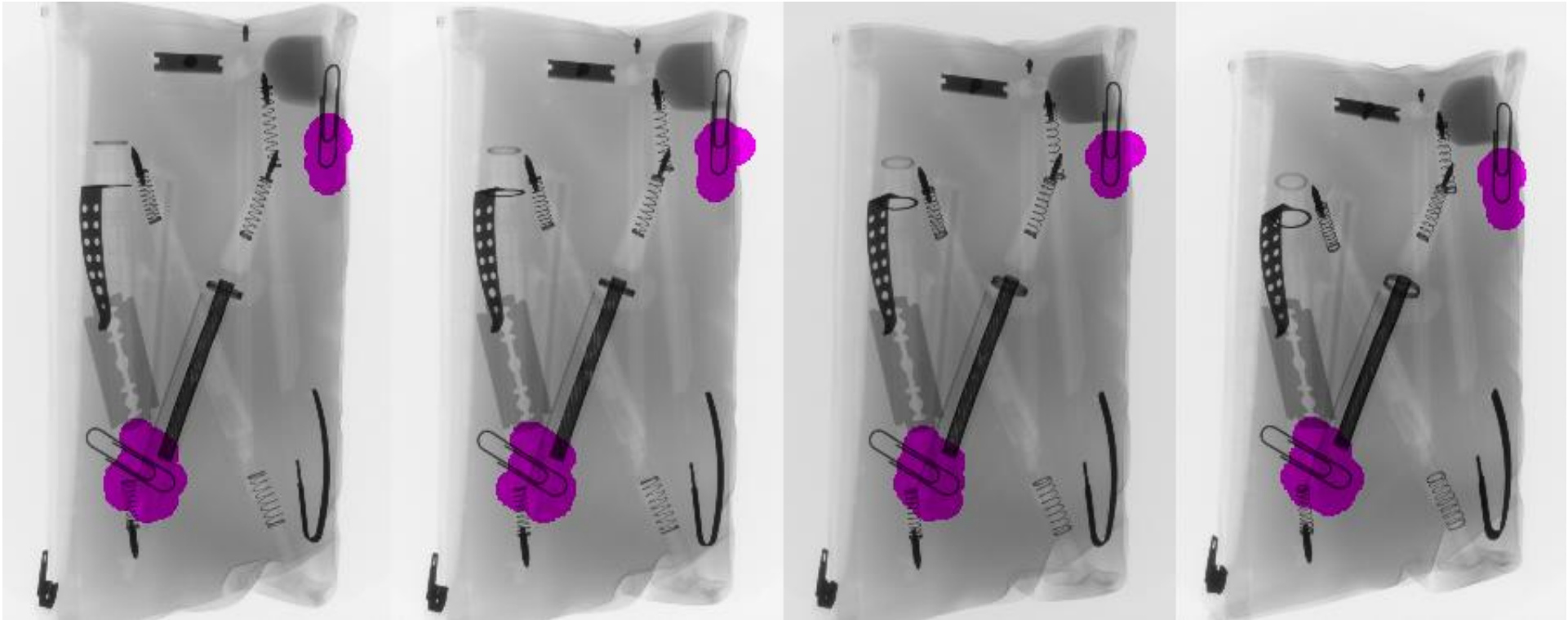


Multiple view analysis: Experimental Results



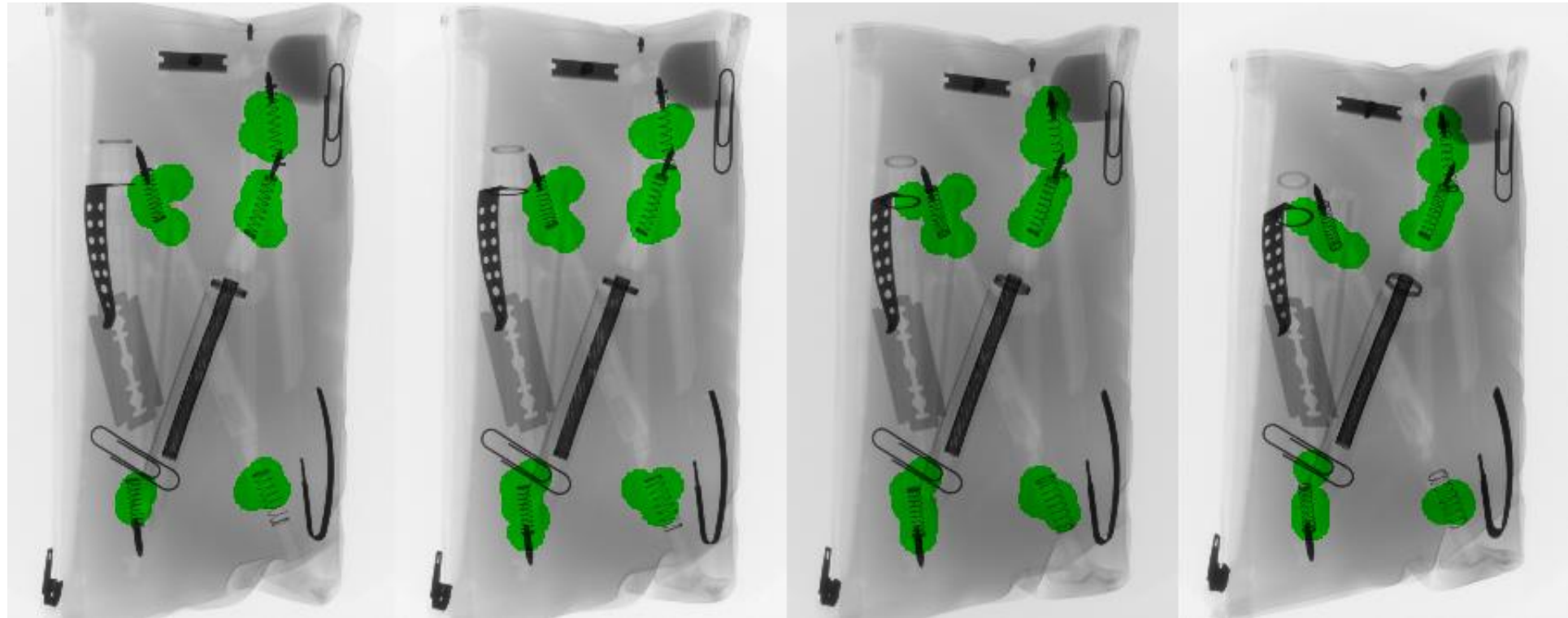
[BLADE]

Multiple view analysis: Experimental Results



[CLIPS]

Multiple view analysis: Experimental Results



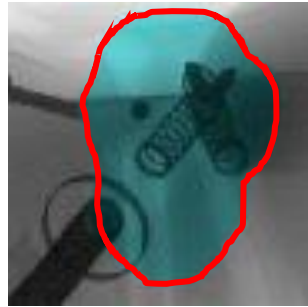
[SPRINGS]

Multiple view analysis: Experimental Results

Recognition with some degree of occlusion



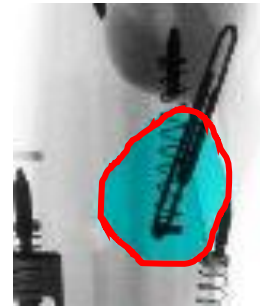
[SPRING]



[SPRINGS]



[CLIP]

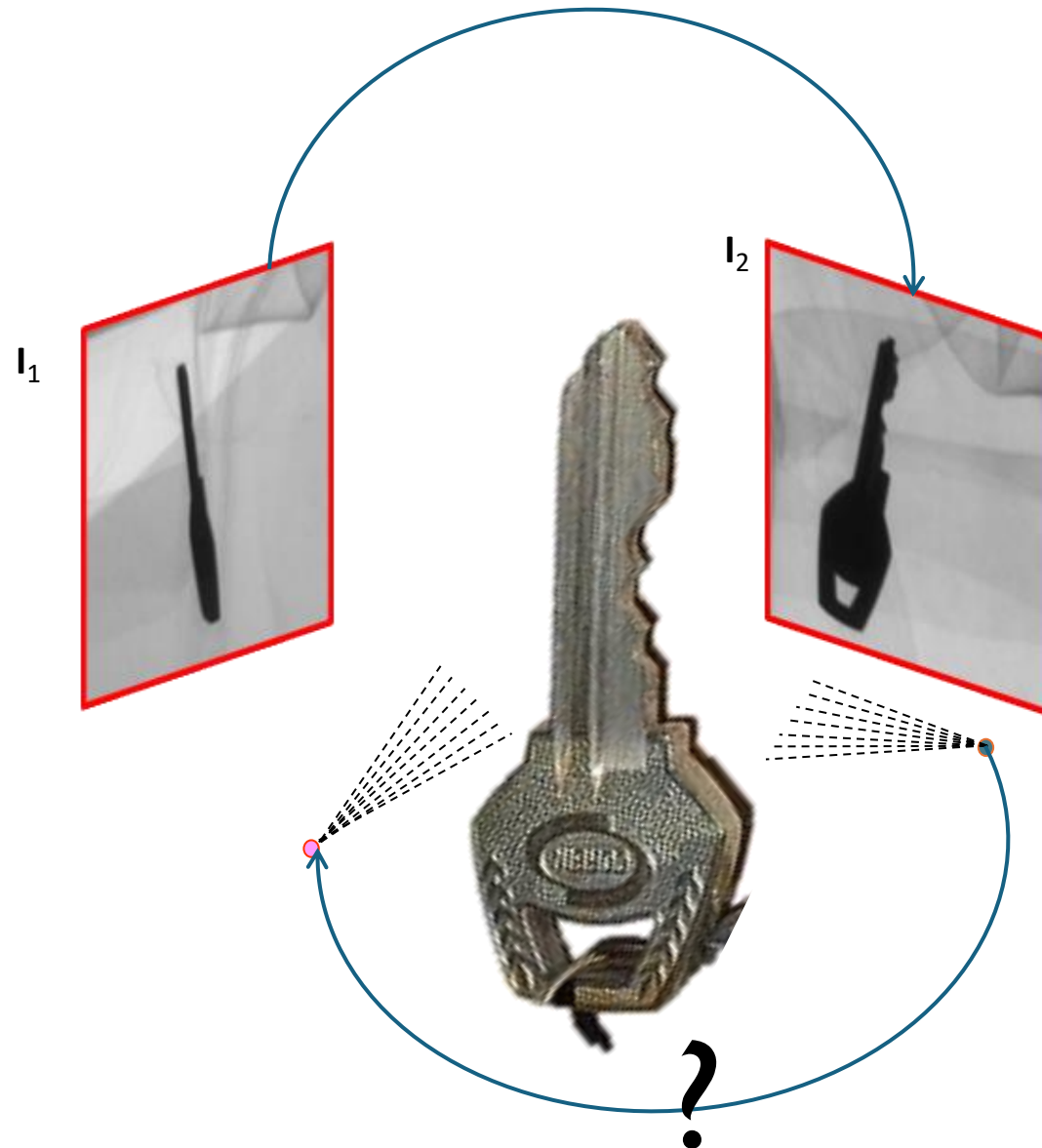


[CLIP]



[BLADE]

Active Vision: Principle



Active Vision: Algorithm

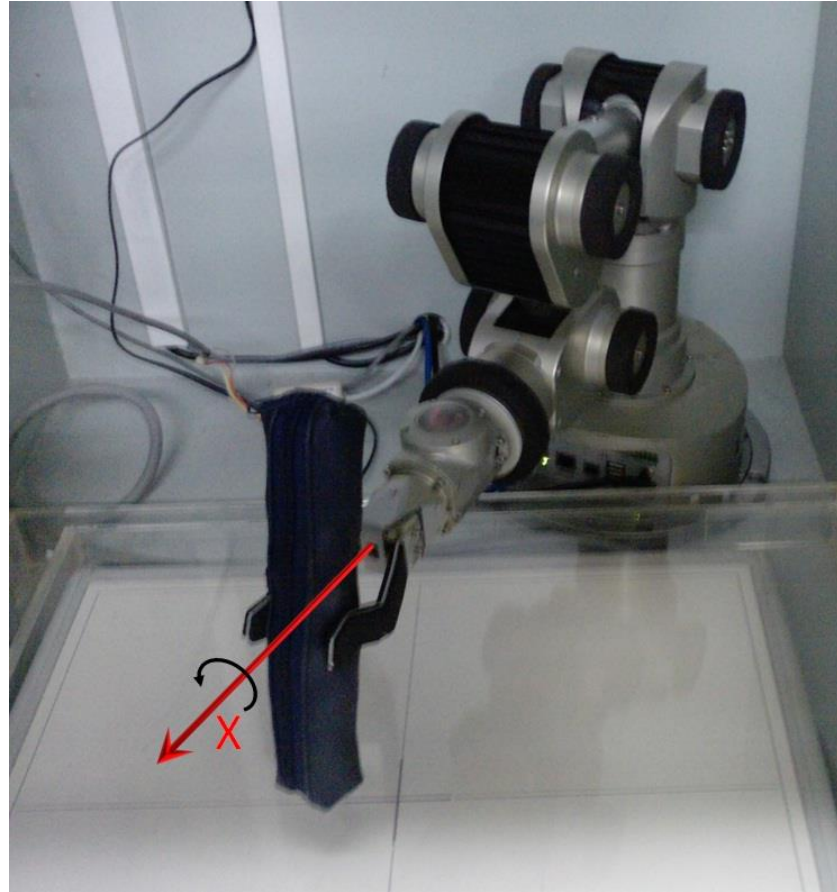
1. Number of Images = 0
Position = Random Position
2. Repeat until Number of Images $< N$
3. Move inspection object to Position
4. Acquisition of a new X-ray image
Number of Images = Number Images + 1
5. Single view detection:

[DETECTION] End

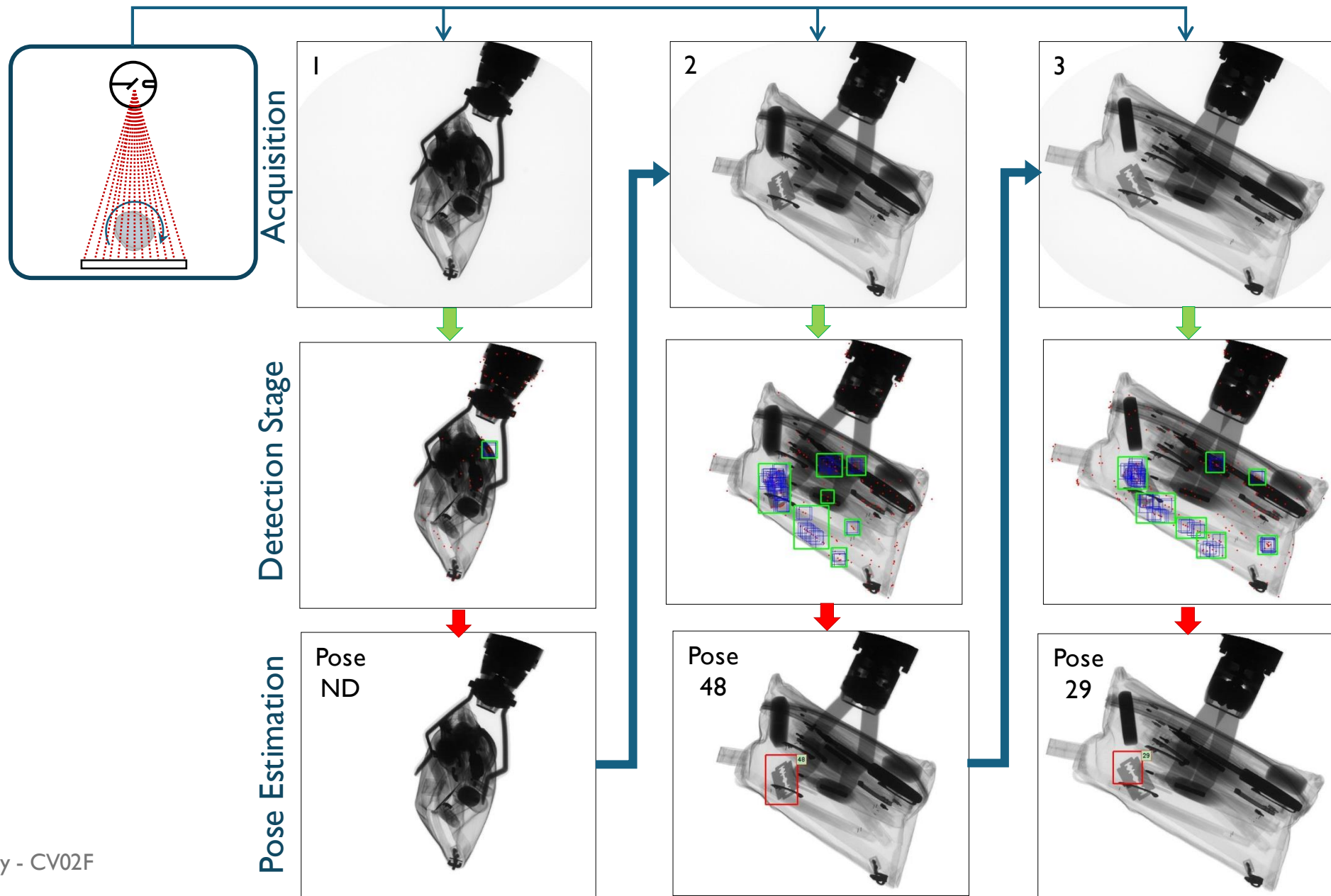
[NOTHING] Position = Position + Θ
Go to step 2

[COULD BE] Pose estimation
Estimation of Better Position
Position = Better Position
Go to step 2

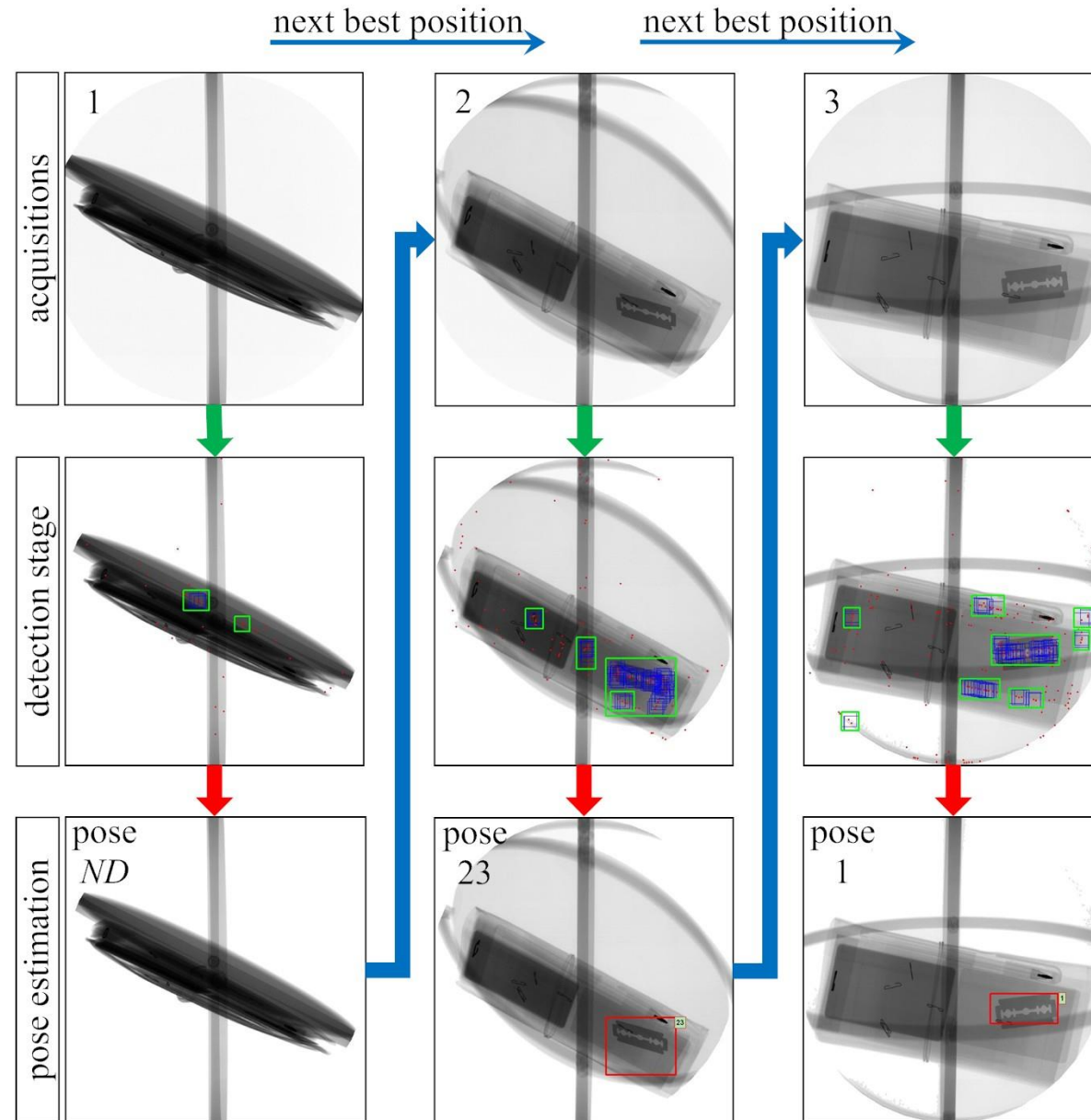
Active Vision: Robotic system



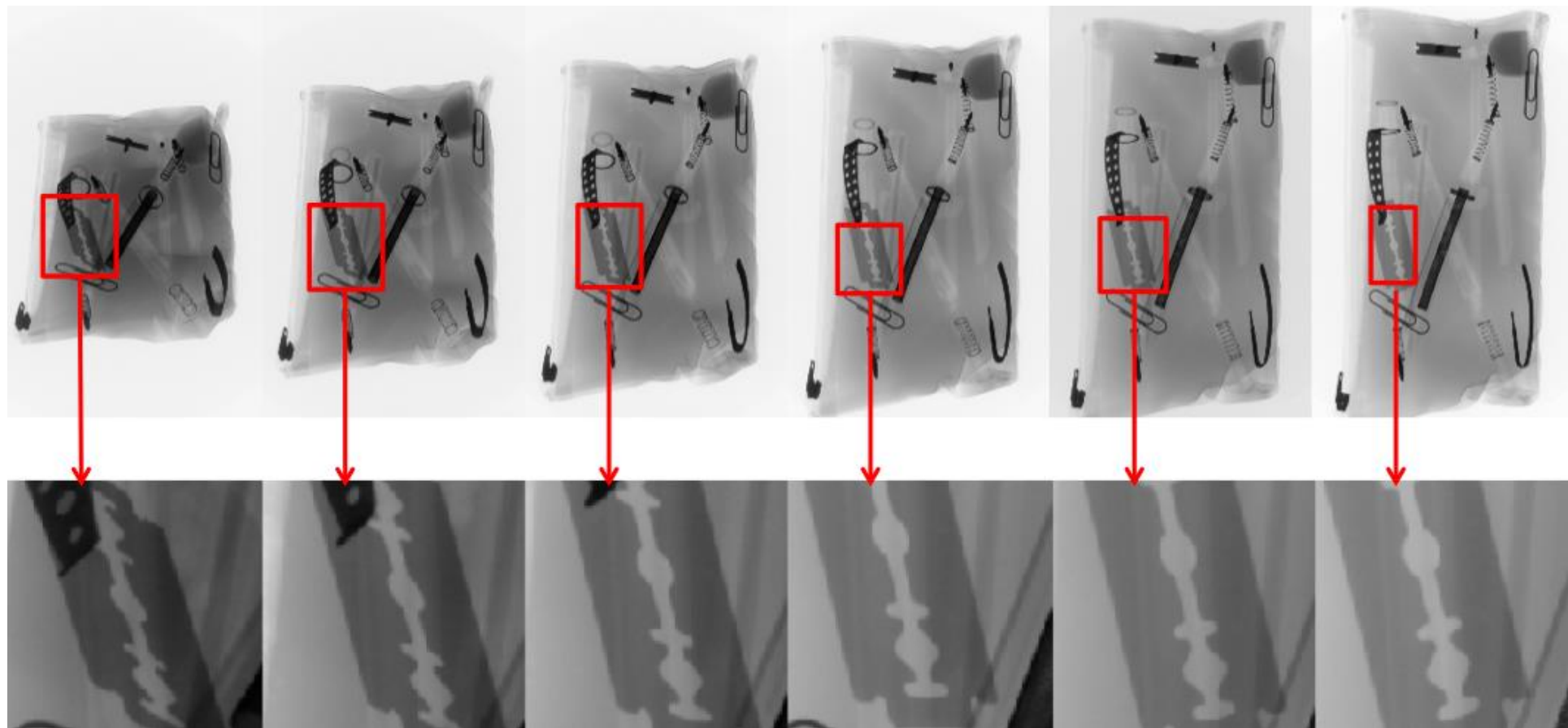
Active Vision: Experimental results



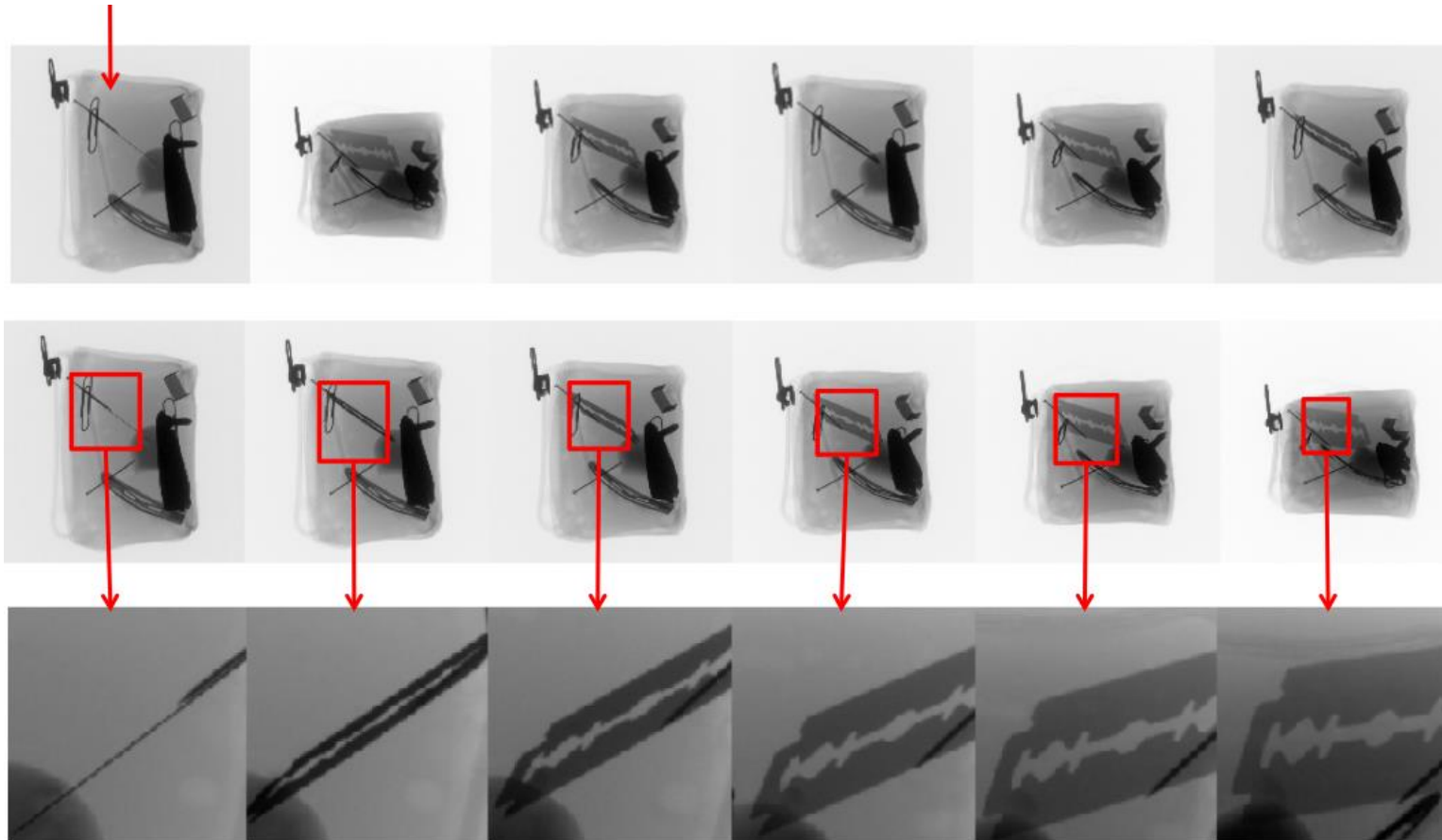
Active Vision: Experimental results



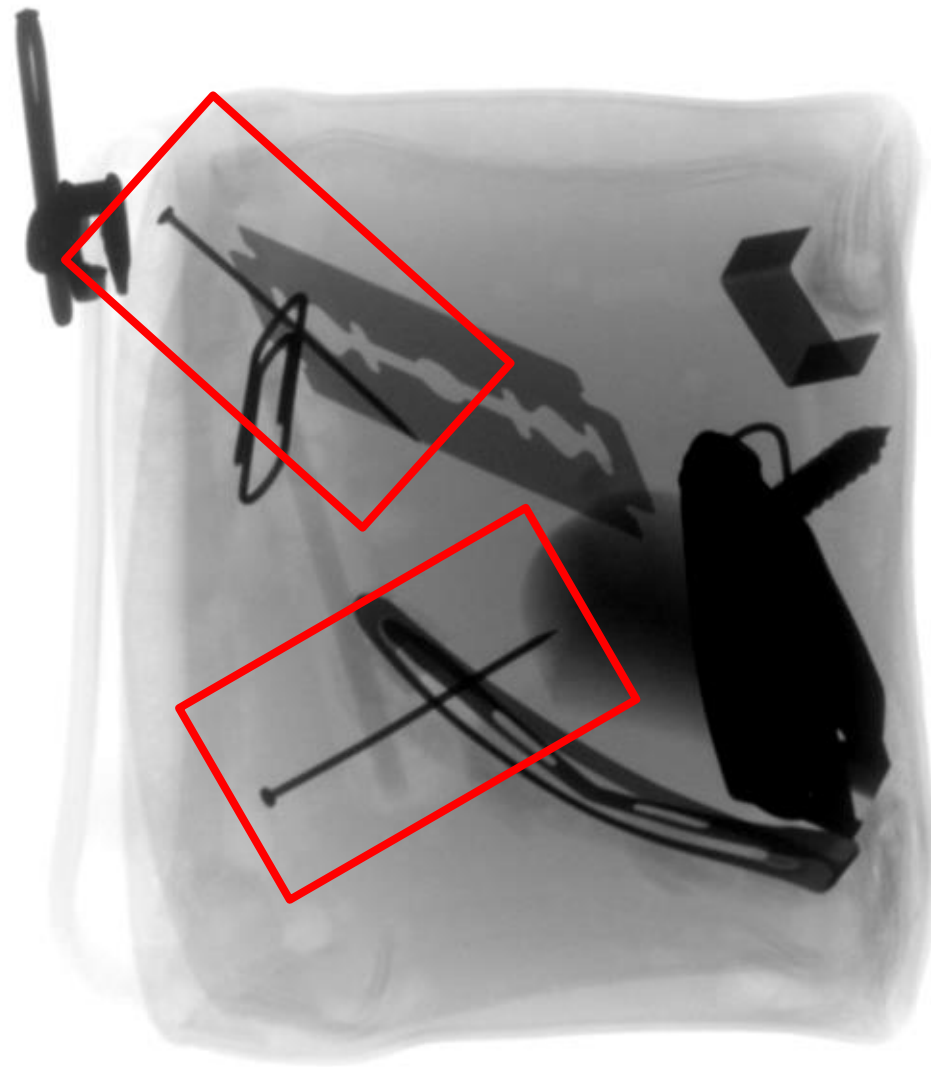
More Examples



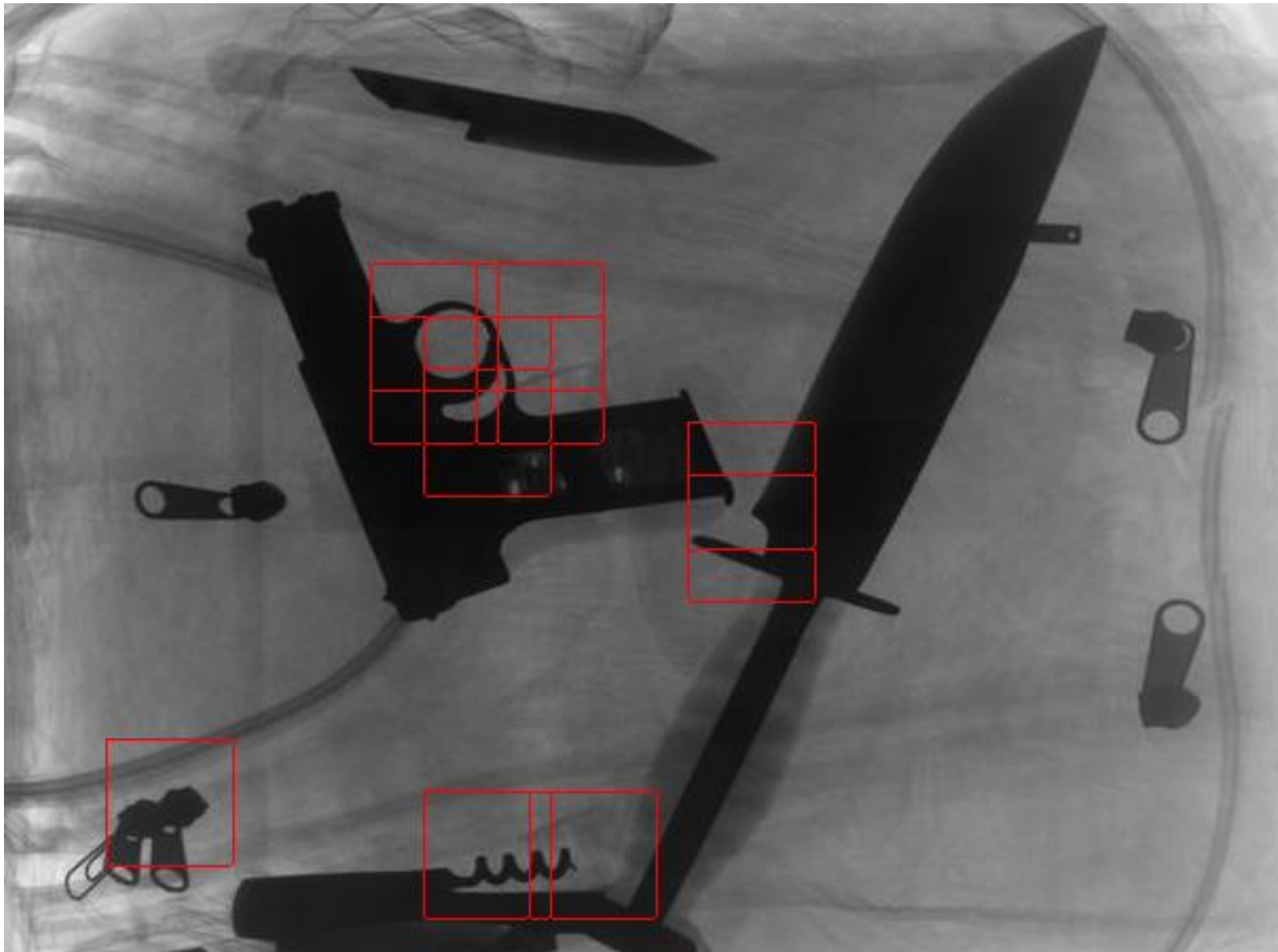
Detection of a razor blade

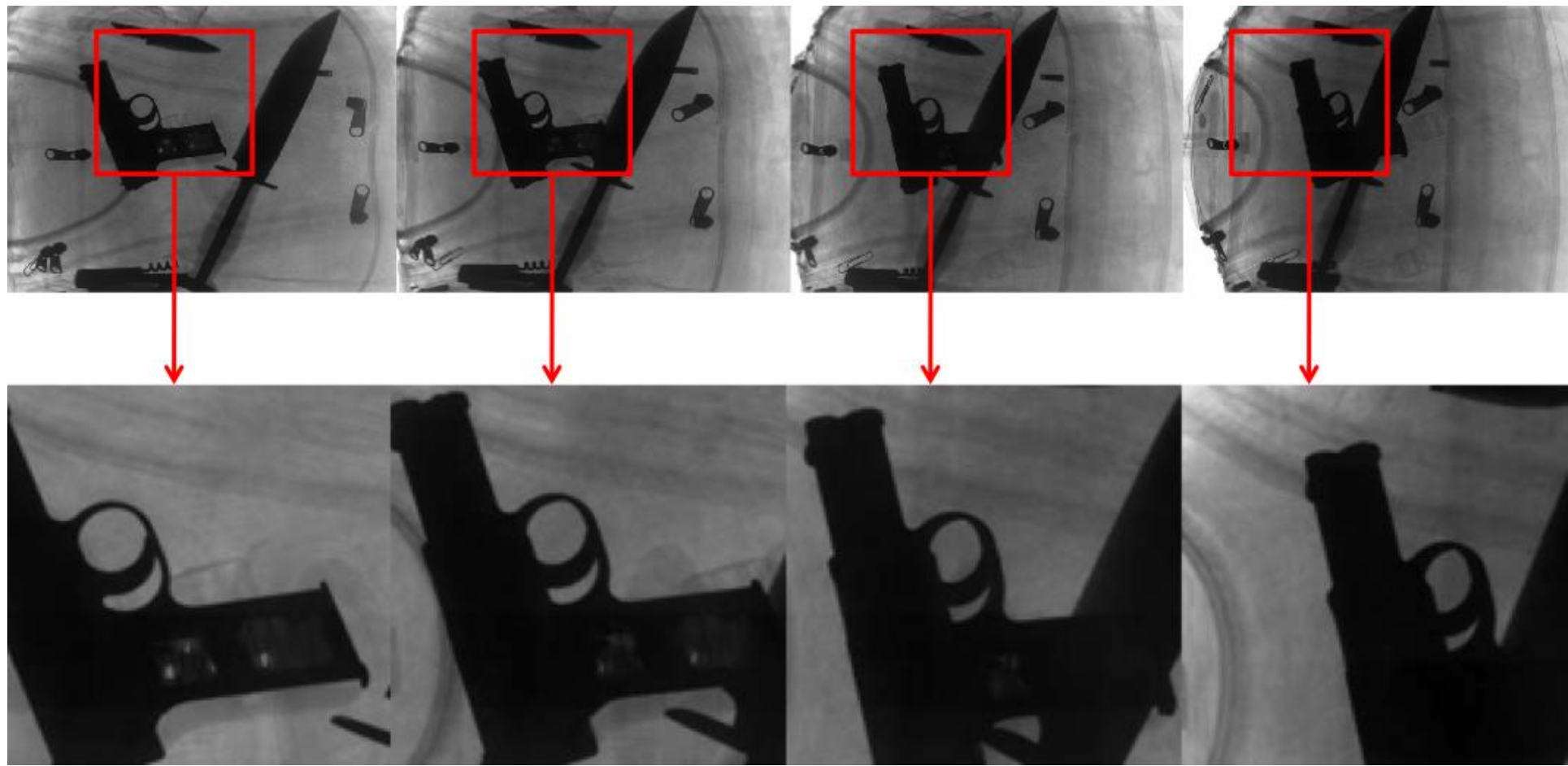


Detection of a razor blade

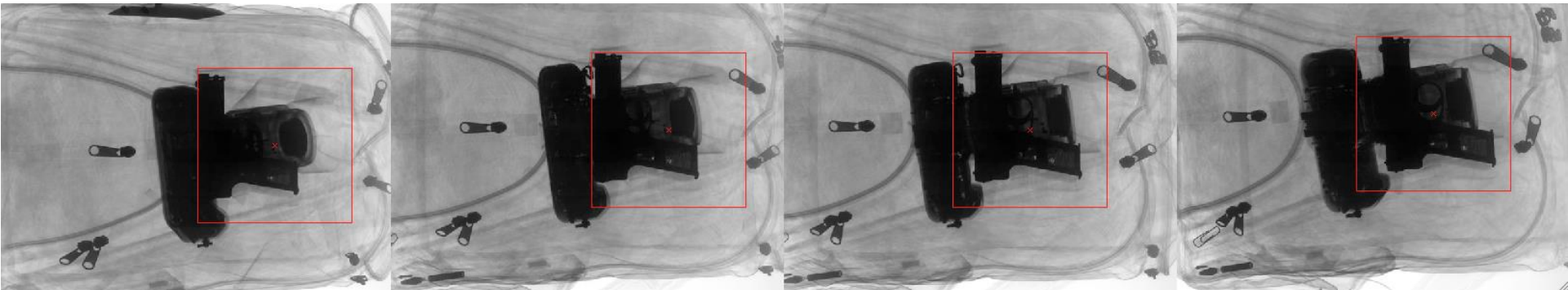


Detection of pins



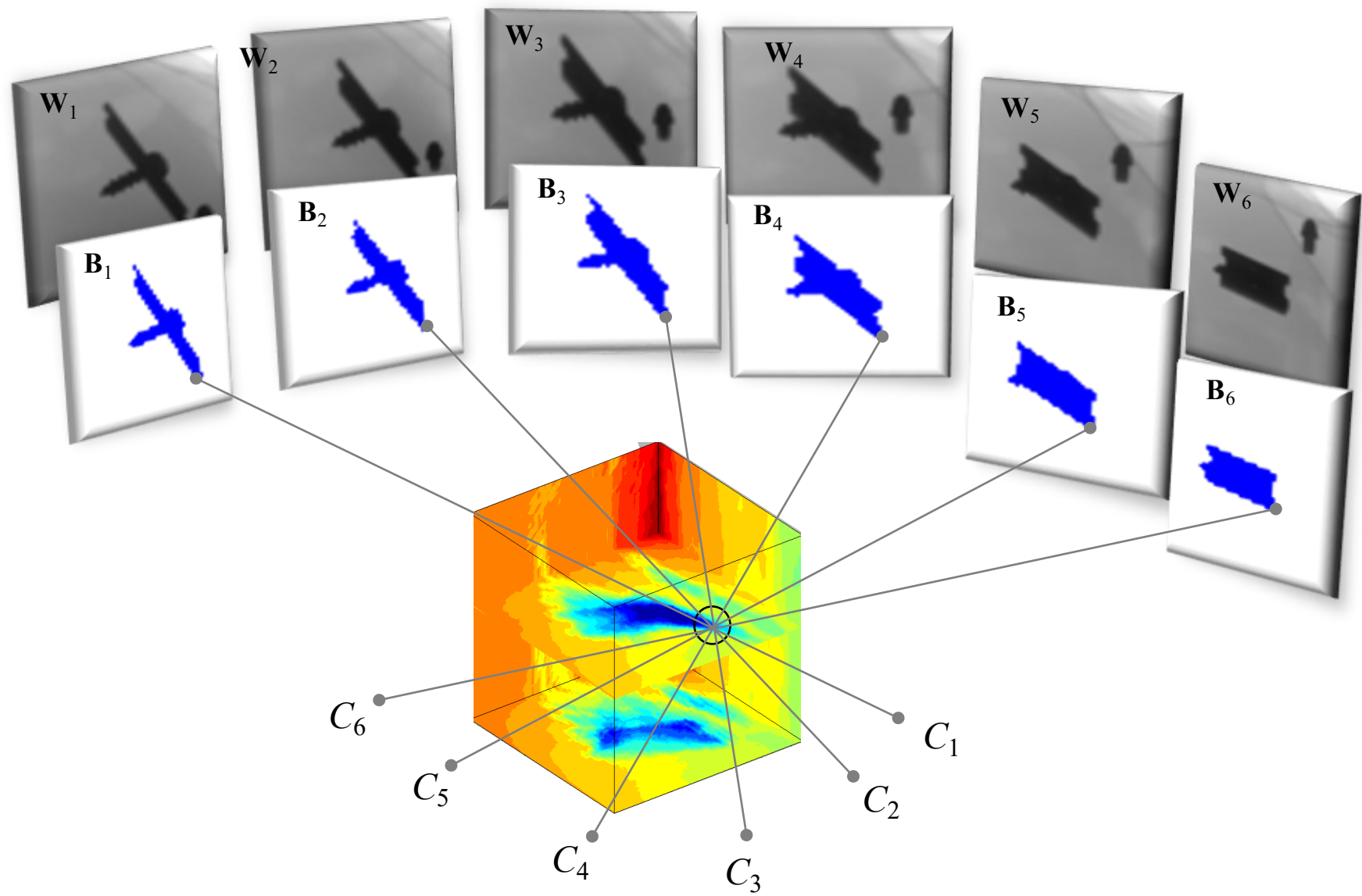


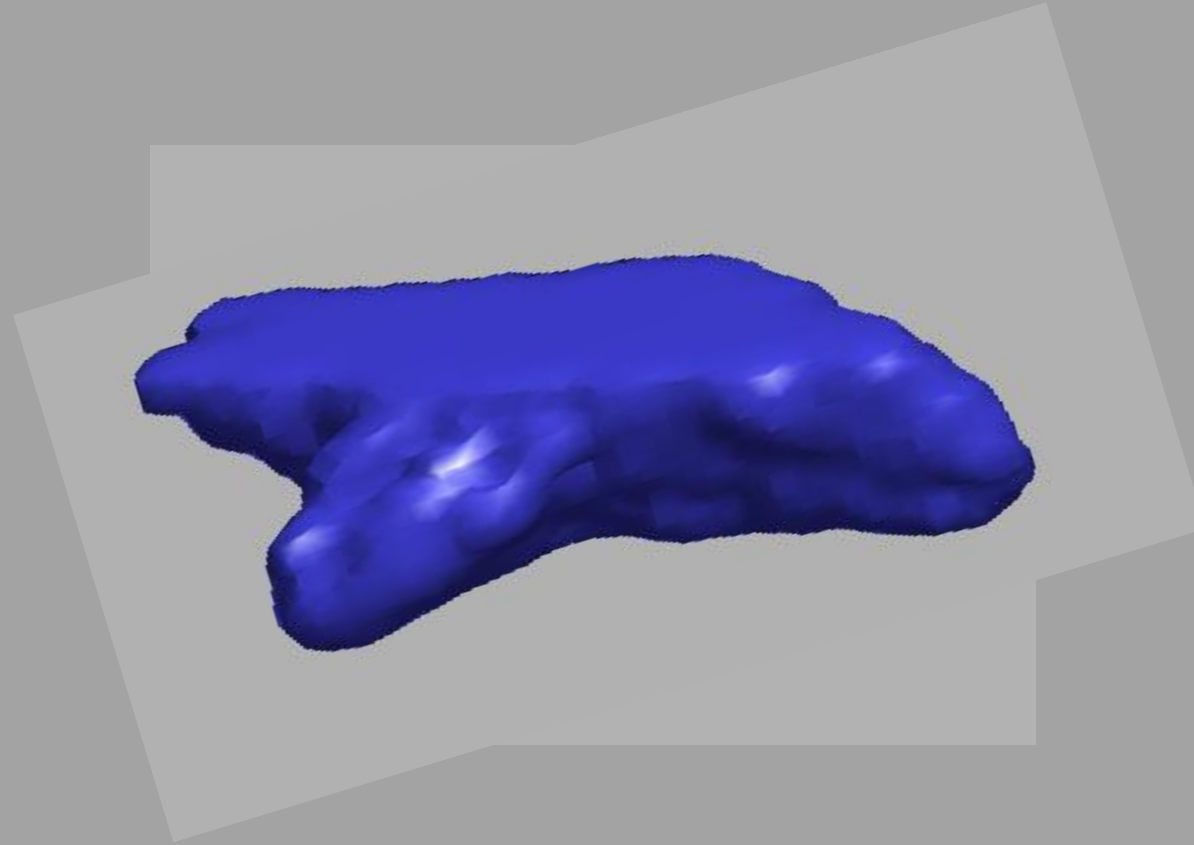
Detection of guns (multiple view)

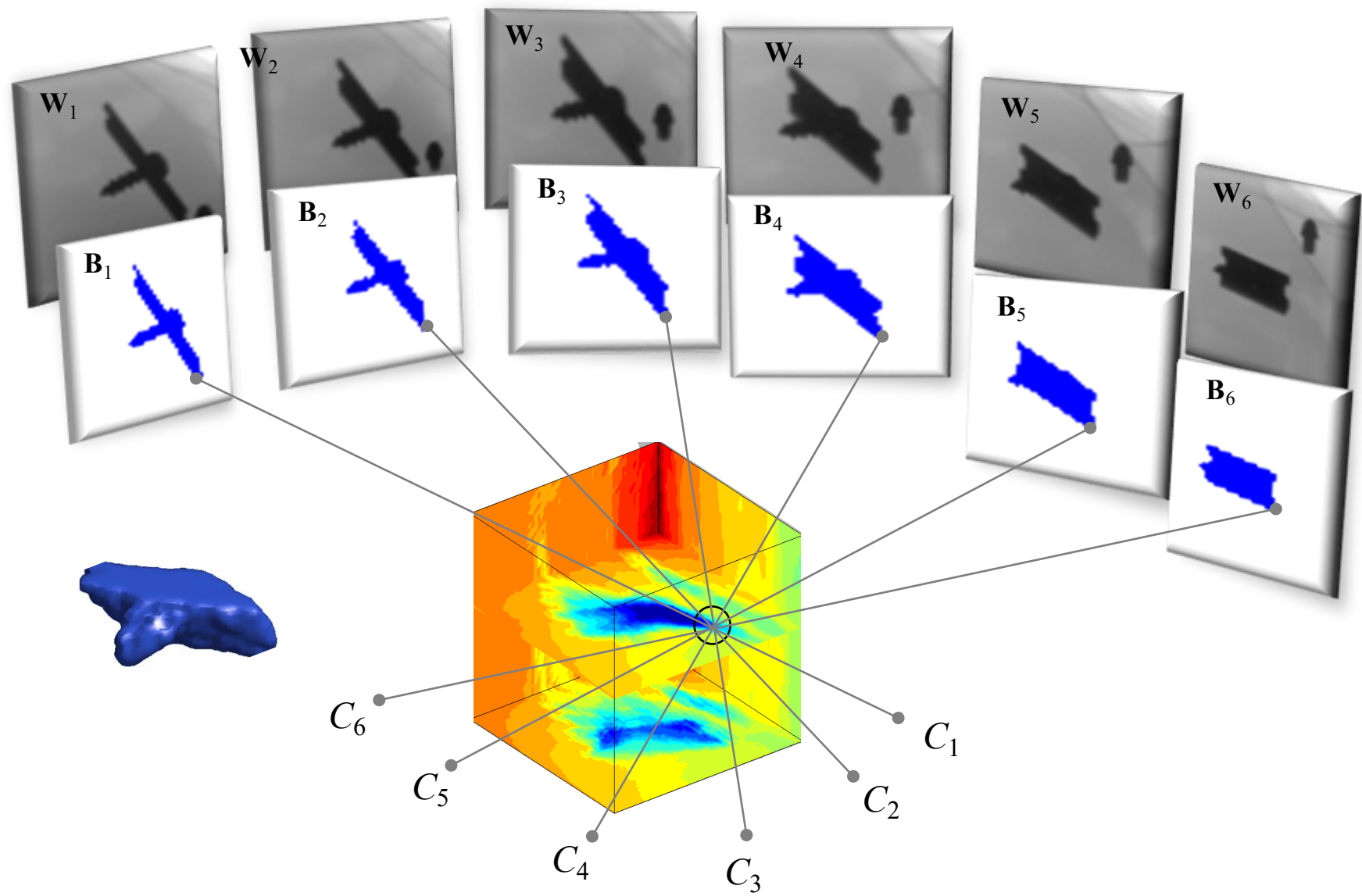


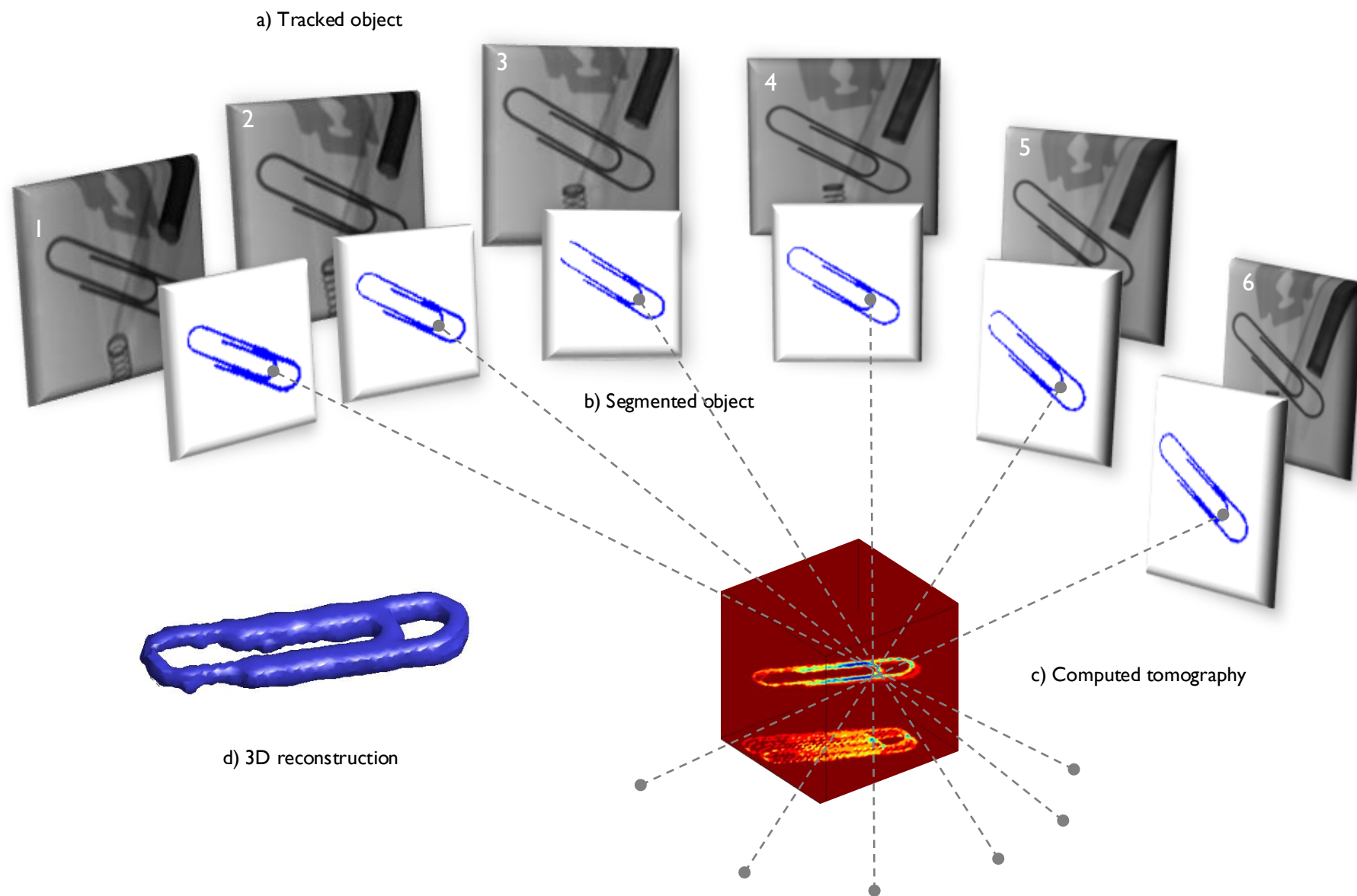
What can we do with a sequence?

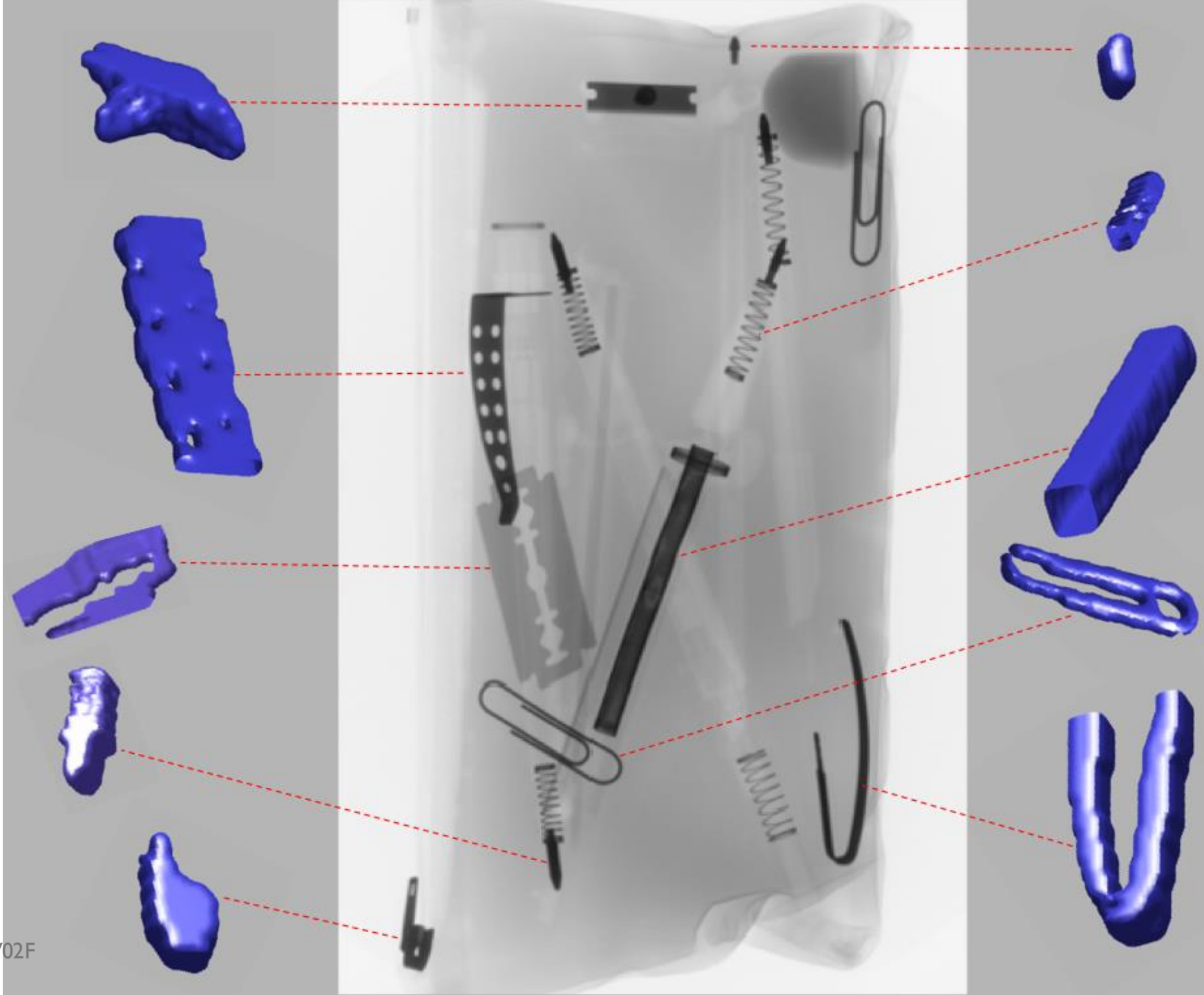












Images

Load seq_guns2.mat

Show 1/4 - +

Object

Search ok

Select 1/2 - +

☒ All objects

☐ One object

☒ All frames

Tracking

None ▾

☐ Structure

☐ Details

☐ Multi

Tomography

Go

Backprojection ▾

X Y Z

slice [64 0 0]

3D [12 4 64]

